
Eliminating Solution Bias in Differentially Private Optimization

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Abstract

Differentially private (DP) stochastic optimization algorithms are widely used in privacy-preserving deep learning, where per-sample gradient clipping and noise injection protect sensitive information. However, these operations limit existing DP algorithms to converge within a constant-radius neighborhood of the first-order stationary point, leading to solution bias and the well-known privacy-utility trade-off. To enhance model utility, we propose a novel algorithmic framework called DP-C4, which is designed to be error-Consistently-vanishing, Coupledly-clipped, solution-Calibrated, and Convergence-guaranteed. Specifically, it incorporates a carefully designed coupled clipping scheme with shifted threshold strategy, ensuring that both clipping bias and noise variance asymptotically vanish, thereby eliminating the DP-induced solution bias. Moreover, we extend existing sensitivity analysis techniques and develop a tailored privacy budget allocation to guarantee the privacy of DP-C4. Compared with the well-recognized DP-SGD, our framework injects significantly less noise under the same privacy level. In addition, we prove that our framework converges to the optimum in strongly-convex case and to a diminishing neighborhood of the first-order stationary point in non-convex case. Experiments show that DP-C4 achieves superior privacy-utility trade-off over existing baselines across various tasks and datasets.

1. Introduction

Background Deep learning have been extensively applied in numerous fields, such as smart homes (Li et al., 2023), transportation (Tahaei et al., 2020), and healthcare (Tang et al., 2019). However, the individual privacy whose information is included in datasets should be protected when the models are actually applied. Therefore, it is important to design privacy-preserving algorithms.

Differential Privacy (DP) (Dwork et al., 2006; Dwork & Roth, 2014) has emerged as the gold standard for privacy-preserving deep learning. It offers provable privacy guarantees that the algorithm learns from sensitive data while limiting the information leaked about any individual sample. To protect the privacy of the training data, numerous differentially private stochastic optimization algorithms have been proposed for deep learning, such as DP stochastic gradient descent (DP-SGD) (Abadi et al., 2016). They apply per-sample gradient clipping and add Gaussian noise to the aggregated gradient, which have been successfully deployed in both centralized (McMahan et al., 2018b; Bu et al., 2020) and federated (Geyer et al., 2017; Truex et al., 2020) settings.

However, the perturbation introduced by gradient clipping and noise often leads to reduced model accuracy. Therefore, these methods face a trade-off between model utility and privacy (Amin et al., 2019; Zhang et al., 2023b; Xiao et al., 2023). This challenge has attracted considerable attention, leading to the development of several improved variants of DP stochastic optimization algorithms. In particular: (1) adaptive clipping thresholds (Andrew et al., 2021; Phan et al., 2017; Pichapati et al., 2019) are adopted to reduce noise variance; (2) gradient normalization or group-based clipping (Yang et al., 2022; Das et al., 2021; McMahan et al., 2018a) are designed to mitigate clipping bias; and (3) iterative schemes are transferred from advanced non-DP optimizers (Zhu et al., 2024; Murata & Suzuki, 2023; Lee, 2017) to leverage their advantageous properties. Nevertheless, gradient clipping and added noise inevitably alter the original optimization dynamics. Prior work shows that under settings similar to DP-SGD, regardless of how the clipping threshold or step size is chosen, DP algorithms only converge with a constant bias term, i.e., converge to

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a neighborhood of the first-order stationary point with a constant radius (Chen et al., 2020; Xiao et al., 2023; Song et al., 2013). Recently, the DiceSGD algorithm (Zhang et al., 2023a) integrates an Error Feedback mechanism to eliminate clipping bias at each iteration, enabling convergence *in expectation over the injected noise*. However, it does not account for noise variance, thereby driving the iterates to drift away from the optimum, leaving the solution bias issue. As a result, existing DP algorithms fail to handle both clipping bias and noise variance. This naturally motivates a fundamental but important question:

Is it possible to design a DP stochastic optimization algorithm that both clipping bias and noise variance asymptotically vanish during iterations, thereby eliminating the issue of solution bias?

Our Contributions We provide an affirmative answer to the question by proposing an **error-Consistently-vanishing, Coupledly-clipped, solution-Calibrated, and Convergence-guaranteed** (DP-C4) algorithmic framework. This framework incorporates a carefully designed coupled clipping scheme and shifted threshold strategy, which enforces the clipping bias and noise variance to asymptotically vanish during iterations, making the convergence point align with the solution of the original optimization problem, thus eliminating the solution bias caused by DP. To the best of our knowledge, this is the first time such a framework is proposed. Furthermore, by closely examining the structure of DP-C4, we have extended the existing privacy analysis tools, thereby providing privacy guarantees for our framework. On the other hand, we prove that our method converges to the optimum in strongly-convex case and to a diminishing neighborhood of the first-order stationary point in the nonconvex case. We conduct extensive experiments to demonstrate the superior utility performance of the DP-C4 framework. Specifically, our contributions are as follows:

- **Utility-Enhanced Framework:** We propose DP-C4, the first DP stochastic optimization algorithmic framework that eliminates solution bias by ensuring the joint asymptotic vanishing of noise variance and clipping bias, thereby achieving a more favorable privacy–utility trade-off.
- **Extended Privacy Analysis:** We extend existing sensitivity analysis techniques and develop a tailored privacy budget allocation strategy to guarantee the privacy of DP-C4. Compared with DP-SGD, our framework injects significantly less noise under the same privacy level. Notably, the proposed sensitivity analysis offers a new perspective for handling differentially private algorithms in which local and published information are intrinsically coupled.
- **Stronger Convergence Guarantee:** We establish the convergence guarantee of DP-C4 by carefully examining its specialized iterative structure. Specifically, this framework converges to the optimum by properly constructing Lyapunov functions in strongly-convex case, and to a diminishing neighborhood of the first-order stationary point in nonconvex case. To the best of our knowledge, this is the first DP algorithm whose convergence admits an unbiased Lyapunov analysis, as a result of an inherent solution calibration effect that mitigates DP-induced solution bias.
- **Empirical Validation:** We conduct extensive experiments showing our framework achieves superior privacy-utility trade-offs over existing baselines across various tasks and datasets.

2. Preliminaries

2.1. Problem Setup and Definitions

Problem Setup We consider the empirical risk minimization (ERM) problem on a dataset D with size $|D| = N$:

$$\min_{x \in \mathbb{R}^d} f(x) := \frac{1}{N} \sum_{i=1}^N f_i(x), \quad (1)$$

where $f_i(x)$ denotes the loss associated with the i -th data sample. Our goal is to propose a DP stochastic optimization algorithmic framework with Gaussian mechanism for finding its first-order stationary point x^* , i.e., $\nabla f(x^*) = \frac{1}{N} \sum_{i=1}^N \nabla f_i(x^*) = 0$.

Differential Privacy We adopt the standard notion of (ϵ, δ) -differential privacy.

Definition 1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006)). A randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ is said to satisfy (ϵ, δ) -DP if for any two neighboring datasets $D, D' \in \mathcal{D}$ differing in at most one data record, and for any measurable subset $\mathcal{S} \subseteq \mathcal{R}$, it holds that

$$\Pr[\mathcal{M}(D) \in \mathcal{S}] \leq e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{S}] + \delta. \quad (2)$$

Here, $\epsilon > 0$ is the privacy budget controlling the strength of privacy protection, and $\delta \in [0, 1]$ denotes a negligible probability of failure.

Sensitivity The noise required by DP is determined by the ℓ_2 -sensitivity of the target function.

Definition 2 (ℓ_2 -Sensitivity). For a function $f : \mathcal{D} \rightarrow \mathbb{R}^d$, its ℓ_2 -sensitivity is defined as

$$\Delta_f := \max_{D, D'} \|f(D) - f(D')\|, \quad (3)$$

where $\|\cdot\|$ denotes ℓ_2 -norm. The maximum is taken over all neighboring datasets D and D' .

Gaussian Mechanism The Gaussian mechanism perturbs the output of a function with additive Gaussian noise calibrated to its sensitivity.

Definition 3 (Gaussian Mechanism (Dwork & Roth, 2014)). Given a function $f : \mathcal{D} \rightarrow \mathbb{R}^d$ and dataset $D \in \mathcal{D}$, the Gaussian mechanism adds noise calibrated to the ℓ_2 -sensitivity of f :

$$\mathcal{M}(D) = f(D) + \mathcal{N}(0, \Delta_f^2 s^2 \mathbf{I}_d), \quad (4)$$

where $\mathcal{N}(0, \Delta_f^2 s^2 \mathbf{I}_d)$ denotes a d -dimensional Gaussian distribution with covariance $\Delta_f^2 s^2 \mathbf{I}_d$, and $s \geq \frac{\sqrt{2 \log(1.25/\delta)}}{\epsilon}$.

2.2. DP-SGD

DP-SGD (Abadi et al., 2016) is a widely adopted algorithmic framework for solving (1) while preserving the privacy of user data. At each iteration, DP-SGD enforces differential privacy by clipping per-sample gradients and injecting Gaussian noise calibrated to the clipping threshold.

Algorithm 1 DP-SGD

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1: Initialize  $x^0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:   Sample  $S_k \subseteq D$ 
4:    $g_i^k = \text{clip}(\nabla f_i(x^k), C)$ 
5:    $\tilde{g}_i^k = g_i^k + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ 
6:    $\tilde{g}^k = \frac{1}{|S_k|} \sum_i \tilde{g}_i^k$ 
7:    $x^{k+1} = x^k - \eta \tilde{g}^k$ 
8: end for
    
```

Vanilla DP-SGD with Fixed Clipping At the k -th iteration, DP-SGD randomly samples a mini-batch $S_k \subseteq D$, clips the ℓ_2 -norm of each stochastic gradient, and then adds Gaussian noise. With a fixed clipping threshold C , the update rule is given by

$$x^{k+1} = x^k - \frac{\eta}{|S_k|} \sum_{i \in S_k} \left(\text{clip}(\nabla f_i(x^k), C) + \mathcal{N}(0, \Delta^2 s^2 \mathbf{I}_d) \right),$$

where $\eta > 0$ is the step size and $\text{clip}(\nabla f_i(x^k), C) := \nabla f_i(x^k) \min \left\{ 1, \frac{C}{\|\nabla f_i(x^k)\|} \right\}$. The clipping threshold C ensures a uniform bound on the sensitivity of the stochastic gradient estimator, which directly determines the scale of the injected noise. Therefore, we denote $\sigma^2 C^2 = \Delta^2 s^2$ for notational convenience in the subsequent analysis. The complete DP-SGD algorithm is presented in Alg.1.

DP-SGD with Threshold Decaying To reduce the magnitude of injected noise as the optimization progresses, a more flexible strategy allows the clipping threshold to vary over iterations, i.e., C_k replaces the fixed constant C .

From a noise-reduction perspective, it is desirable to set $C_k \rightarrow 0$ as $x^k \rightarrow x^*$, so that the variance of the injected

Gaussian noise vanishes near the optimum. However, in stochastic optimization, the variance of mini-batch gradients generally does not vanish at the optimum. In particular, even when $x^k \rightarrow x^*$, we typically have $\left\| \frac{1}{|S_k|} \sum_{i \in S_k} \nabla f_i(x^k) \right\| \neq \left\| \frac{1}{|D|} \sum_{i \in D} \nabla f_i(x^*) \right\| = 0$. As a consequence, even in a neighborhood of the optimal point, setting C_k too small leads to persistent gradient clipping, which still introduces a non-negligible bias into the gradient estimator and prevents convergence to the optimum.

Limitation In summary, existing DP-framework design suffers from an inherent trade-off between noise variance and clipping bias: A large clipping threshold reduces bias but requires injecting substantial noise to maintain privacy, whereas a small threshold suppresses noise at the cost of excessive clipping bias. This tension fundamentally limits the ability of existing schemes.

3. Framework: DP-C4

In this section, to overcome the aforementioned limitation, we propose a novel DP stochastic optimization framework called **DP-C4**, which is **error-Consistently-vanishing, Coupledly-clipped, solution-Calibrated, and Convergence-guaranteed**. This framework ensures the asymptotic vanish of both the noise variance and the clipping bias, thereby eliminating solution bias.

3.1. High-level Idea

We begin by investigating the forms of iterative scheme in which both types of errors vanish asymptotically. To do this, we consider a general setting where the gradient estimator is aggregated from multiple sub-estimators $\{h^{(j)}(x)\}_{j \in [n]}$ satisfying $\sum_{j \in [n]} \mathbb{E}[h^{(j)}(x)] = \nabla f(x)$. Each sub-estimator is defined by $h^{(j)}(x) := \frac{1}{|S_j|} \sum_{i \in S_j} h_i^{(j)}(x)$, where $S_j \subseteq D$ denotes the sampled dataset and $\{h_i^{(j)}\}_{i \in S_j, j \in [n]}$ denotes per-sample estimators. Furthermore, for DP algorithms, we clip the ℓ_2 norm of each component $h_i^{(j)}(x)$, aggregate the clipped components and add noise to form the DP gradient estimator $\tilde{g}^k$. For simplicity, we focus on the case $n = 2$. The iterative scheme is given by:

$$\begin{cases} x^{k+1} = x^k - \eta \tilde{g}^k, \\ \tilde{g}^k = \left[\frac{1}{|S_1|} \sum_{i \in S_1} \text{clip}(h_i^{(1)}(x^k), C_1) + \mathbf{n}_1^k \right] \\ \quad + \left[\frac{1}{|S_2|} \sum_{i \in S_2} \text{clip}(h_i^{(2)}(x^k), C_2) + \mathbf{n}_2^k \right], \\ \mathbf{n}_1^k \sim \mathcal{N}(0, \sigma_1^2 C_1^2 \mathbf{I}), \quad \mathbf{n}_2^k \sim \mathcal{N}(0, \sigma_2^2 C_2^2 \mathbf{I}). \end{cases}$$

Here, C_j is the clipping threshold and σ_j^2 is the privacy-dependent noise multiplier. Instead of using fixed C_j during

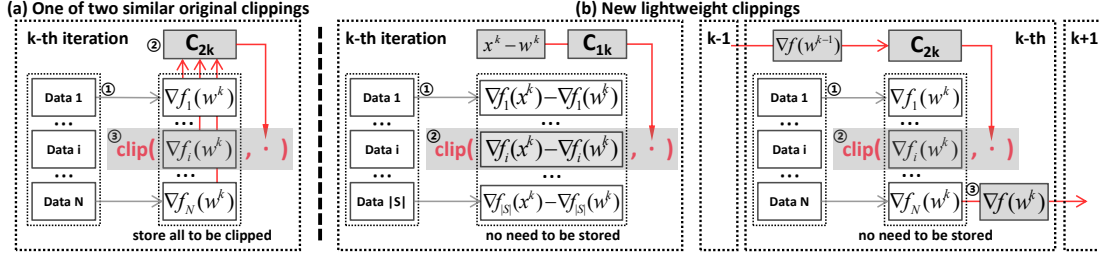


Figure 1. Workflow of Two Different Kinds of Clippings.

iterations, we consider replacing them with an estimator-dependent function $C_j(\{h_i^{(j)}(x^k)\}_{i \in S_j})$, forcing both clipping bias $B_k^{(j)}$ and noise variance $V_k^{(j)}$ vanish as $x^k \rightarrow x^*$:

$$\begin{cases} B_k^{(j)} := \left\| \frac{1}{|S_j|} \sum_{i \in S_j} \text{clip}(h_i^{(j)}(x^k), C_j) - \frac{1}{|S_j|} \sum_{i \in S_j} h_i^{(j)}(x^k) \right\|^2 \\ \quad \xrightarrow{x^k \rightarrow x^*} 0, \\ V_k^{(j)} := \sigma_j^2 C_j^2(\{h_i^{(j)}(x^k)\}) \xrightarrow{x^k \rightarrow x^*} 0. \end{cases}$$

To guide the design, we first establish an upper bound on clipping bias in Lemma.1 (proof in Appendix.D):

Lemma 1 (Upper Bound on Clipping Bias). *Let $I_1^k := \{i \in S : \|h_i(x^k)\| < C(\{h_i(x^k)\}_{i \in S})\}$ be the set of unclipped samples, and $I_2^k := \{i \in S : \|h_i(x^k)\| \geq C(\{h_i(x^k)\}_{i \in S})\}$ the clipped ones. Then,*

$$B_k \leq \frac{|I_2^k|}{|S|^2} \sum_{i \in I_2^k} \left[\|h_i(x^k)\| - C(\{h_i(x^k)\}_{i \in S}) \right]^2.$$

Lemma.1 implies $B_k \rightarrow 0$ as both $\|h_i(x^k)\| \rightarrow 0$ and $C(\{h_i(x^k)\}_{i \in S}) \rightarrow 0$. Therefore, to push the clipping bias $B_k^{(1)}$ to zero as $x^k \rightarrow x^*$, as a natural choice, we set $\{h_i^{(1)}\}_{i \in S_1}$ and $C_1(\{h_i^{(1)}\}_{i \in S_1})$ as:

$$\begin{cases} h_i^{(1)}(x^k) := \nabla f_i(x^k) - \nabla f_i(x^*), \\ C_1(\{h_i^{(1)}(x^k)\}_{i \in S_1}) := C_1 \cdot \left\| \frac{1}{|S_1|} \sum_{i \in S_1} (\nabla f_i(x^k) - \nabla f_i(x^*)) \right\|, \end{cases}$$

where C_1 is a scaling factor. However, since x^* is unknown, we replace x^* with a history iterate $w^k \in \{x^t\}_{t \in [k]}$:

$$\begin{cases} h_i^{(1)}(x^k, w^k) := \nabla f_i(x^k) - \nabla f_i(w^k), \\ C_1(\{h_i^{(1)}(x^k, w^k)\}_{i \in S_1}) := C_1 \cdot \left\| \frac{1}{|S_1|} \sum_{i \in S_1} (\nabla f_i(x^k) - \nabla f_i(w^k)) \right\|. \end{cases}$$

When $x^k, w^k \rightarrow x^*$, we have $B_k^{(1)} \rightarrow 0$. Meanwhile, $V_k^{(1)} \rightarrow 0$ since $C_1(\{h_i^{(1)}\}_{i \in S_1}) \rightarrow 0$. For the sub-estimator $h^{(2)} = \frac{1}{|S_2|} \sum_{i \in S_2} h_i^{(2)}$, we choose $S_2 = D$, set $\{h_i^{(2)}\}_{i \in S_2}$ and $C_2(\{h_i^{(2)}\}_{i \in S_2})$ as:

$$\begin{cases} h_i^{(2)}(x^k, w^k) := \nabla f_i(w^k), \\ C_2(\{h_i^{(2)}(x^k, w^k)\}_{i \in S_2}) := C_2 \cdot \|\nabla f(w^k)\|, \end{cases}$$

where C_2 is a scaling factor. This choice ensures $\mathbb{E}[h^{(1)}(x^k, w^k) + h^{(2)}(x^k, w^k)] = \nabla f(x^k)$, and makes:

$$B_k^{(2)} + V_k^{(2)} \leq (\sigma_2^2 + 1) \cdot C_2^2 \cdot \|\nabla f(w^k)\|^2 \rightarrow 0 \quad \text{as } w^k \rightarrow x^*.$$

As a result, this design ensures that all error components asymptotically vanish, which forms the foundation of our DP algorithmic scheme:

$$\begin{cases} g^k &= h_1^k + h_2^k \\ h_1^k &= \underbrace{\frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k})}_{\text{coupled gradient difference}} \\ h_2^k &= \underbrace{\frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_{2k})}_{\text{anchor gradient}} \\ C_{1k} &= C_1 \cdot \left\| \frac{1}{|S|} \sum_{i \in S} (\nabla f_i(x^k) - \nabla f_i(w^k)) \right\| \\ C_{2k} &= C_2 \cdot \|\nabla f(w^k)\| \\ w^k &\in \{x^t\}_{t \in [k]} \end{cases} \quad (5)$$

3.2. Practical Threshold Design: Sensitivity Control and Memory Efficiency

While the scheme derived in the previous subsection provide a principled way to eliminate both clipping bias and noise variance asymptotically, directly implementing $C_{1k} = C_1 \cdot \left\| \frac{1}{|S|} \sum_{i \in S_1} (\nabla f_i(x^k) - \nabla f_i(w^k)) \right\|$, $C_{2k} = C_2 \cdot \|\nabla f(w^k)\|$ faces two critical practical challenges:

Issue 1: Sensitivity Mismatch Thresholds in (5) are *current-iteration-dependent*, which breaks the proportionality between the sensitivity of $h_{(i)}^k$ and thresholds themselves (see (28) and (29) in Appendix.C.2 for more details). As a result, the Gaussian mechanism can no longer be calibrated using $C_{(i)k}$, invalidating privacy accounting.

Issue 2: Excessive Memory Cost Moreover, threshold computation requires aggregating per-sample gradients prior to clipping, incurring non-negligible auxiliary memory overhead that is prohibitive for large-scale models.

To address both issues, note that the gradient difference $\|\nabla f_i(x^k) - \nabla f_i(w^k)\|$ can be bounded by $L\|x^k - w^k\|$ under

Algorithm 2 DP-C4

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1: Input: Dataset  $D$ , learning rate  $\eta$ , clipping bounds  $C, C_1, C_2$ , noise scales  $\sigma_1, \sigma_2$ , total steps  $T$ , update probability  $p$ .
2: Output: Model parameters  $x^T$  satisfying  $(\epsilon, \delta)$ -DP.
3: Initialize:  $x^0 = w^0 \in \mathbb{R}^d$ , let  $C_{2(0)} := \min(C, C_2), C_{1(0)} := 0$ .
4: for  $k = 0$  to  $T - 1$  do
5:   Sample  $S \subseteq D$ 
6:    $g_1^k \leftarrow \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k})$  {Coupled term}
7:    $g_2^k \leftarrow \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_{2k})$  {Anchor term}
8:    $C_{2(k+1)} \leftarrow \min(C, C_2 \|\nabla f(w^k)\|)$  {Pre-computed anchor threshold}
9:    $\mathbf{n}_1^k \sim \mathcal{N}(0, \sigma_1^2 C_{1k}^2 \mathbf{I}), \mathbf{n}_2^k \sim \mathcal{N}(0, \sigma_2^2 C_{2k}^2 \mathbf{I})$  {Sample DP noise}
10:   $\tilde{g}^k \leftarrow g_1^k + g_2^k + \mathbf{n}_1^k + \mathbf{n}_2^k$  {Add noise}
11:   $x^{k+1} \leftarrow x^k - \eta \cdot \tilde{g}^k$  {Update model}
12:   $w^{k+1} \leftarrow \begin{cases} x^k, & \text{with probability } p \\ w^k, & \text{with probability } 1 - p \end{cases}$  {Update anchor (Routine 1)}
13:   $C_{1(k+1)} \leftarrow \min(C, C_1 \|x^{k+1} - w^{k+1}\|)$  {Pre-computed coupled threshold}
14: end for
    
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the L -smoothness assumption, which tends to 0 as $x^k, w^k \rightarrow x^*$. Furthermore, rather than using w^k to determine C_{2k} , we shift the clipping and consider using the previous iterate w^{k-1} , which leads to C_{2k} can be computed and stored in advance. Specifically, we make the following substitutions:

$$\begin{cases} C_{1k} = C_1 \cdot \|x^k - w^k\| \\ C_{2k} = \begin{cases} C_2 \cdot \|\nabla f(w^{k-1})\|, & k > 0 \\ C_2, & k = 0 \end{cases} \end{cases}$$

These substitutions resolve both issues simultaneously: The clipping thresholds no longer depend on current mini-batch gradient aggregates, but only on iterate differences and a precomputed full-gradient norm. On the one hand, thresholds can be fixed before per-sample gradient processing, eliminating auxiliary gradient storage and enabling a one-pass, memory-efficient implementation (see Fig.1). On the other hand, current-iteration-independent thresholds can be treated as constants in our extended sensitivity analysis, which ensures that the required noise variance remains linearly bounded by the clipping thresholds, restoring valid noise calibration under the Gaussian mechanism. The extended sensitivity analysis and privacy guarantees are provided in Section.4.1, with a detailed comparative analysis of privacy before and after the substitution in Appendix.C. We emphasize that the substituted clipping thresholds also preserve the key asymptotic properties established in Section.3.1, supplementary discussions are in Appendix.A.

3.3. DP-C4 Algorithm

In this subsection, we formally describe the DP-C4 in Alg.2. Based on the idea in Section.3.1, Alg.2 constructs a gradient estimator by aggregating two sub-estimators: a coupledly-clipped gradient difference term (Line 6) and an anchor term

(Line 7). Specifically, we initialize with $x^0 = w^0 \in \mathbb{R}^d$. At the k -th iteration, we sample a mini-batch $S \subseteq D$ (Line 5). We utilize truncated thresholds using the current iterates and the gradient norm computed in the previous iteration (Line 8,13) to compute the clipped-and-noised gradient estimator $\tilde{g}^k$ (Line 6-7,9-10). For the anchor iterate w^k , we adopt a probabilistic update scheme (Line 12), which implies that both C_{2k} and g_2^k are computed only with a low probability, thereby significantly reducing the overall computational cost. Moreover, there are also alternative routines for updating the anchor w^k (see the following Routine 2-4):

$$\begin{cases} w_{R_2}^{k+1} = \begin{cases} x^k, & k = 1(\text{mod } \lceil 1/p \rceil) \\ w^k, & k \neq 1(\text{mod } \lceil 1/p \rceil) \end{cases}, \\ w_{R_3}^{k+1} = \begin{cases} x^{k+1}, & \text{with } p \\ w^k, & \text{with } 1 - p \end{cases}, \\ w_{R_4}^{k+1} = \begin{cases} x^{k+1}, & k = 1(\text{mod } \lceil 1/p \rceil) \\ w^k, & k \neq 1(\text{mod } \lceil 1/p \rceil) \end{cases}. \end{cases} \quad (6)$$

In practical deployment, to further reduce computational overhead, we often select a large batch size $|D'| \gg |S|$ instead of the full dataset size $|D|$ as the anchor batch. This choice also helps to reduce the solution bias and improve utility, since it often holds that $\|\frac{1}{|D'|} \sum_{i \in D'} \nabla f_i(x^*)\| \ll \|\frac{1}{|S|} \sum_{i \in S} \nabla f_i(x^*)\|$.

3.4. Solution-Calibrated Property of DP-C4

In this subsection, to maximally highlight the fundamental distinction of DP-C4 framework, we examine the relationship between its potential convergence points and the ERM solution x^* , where existing DP algorithmic frameworks generally suffer from a constant-order bias between the two. We consider all sources of randomness (i.e., sampling, noising,

and anchor-updating) and investigate from the perspective of fixed-point analysis. Specifically, at a fixed point $(\tilde{x}, \tilde{w})$, both sequences $\{x^k\}_{k \in \mathbb{N}}$ and $\{w^k\}_{k \in \mathbb{N}}$ converge, implying $x^{k+1} = x^k = \tilde{x}$, $w^{k+1} = w^k = \tilde{w}$. Hence, according to Alg.2, the fixed point of DP-C4 satisfies the following system:

$$\begin{cases} \tilde{g}^k = \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(\tilde{x}) - \nabla f_i(\tilde{w}), C_{1k}) \\ \quad + \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(\tilde{w}), C_{2k}) + \mathbf{n}_1^k + \mathbf{n}_2^k = 0, \\ \tilde{w} = w^{k+1} = \begin{cases} x^k = \tilde{x}, & \text{with probability } p, \\ w^k, & \text{with probability } 1 - p, \end{cases} \\ \mathbf{n}_1^k \sim \mathcal{N}(0, \sigma_1^2 C_{1k}^2 \mathbf{I}), \quad \mathbf{n}_2^k \sim \mathcal{N}(0, \sigma_2^2 C_{2k}^2 \mathbf{I}), \\ C_{1k} = \min(C, C_1 \|\tilde{x} - \tilde{w}\|), \\ C_{2k} = \min(C, C_2 \|\frac{1}{|D|} \sum_{i=1}^{|D|} \nabla f_i(\tilde{w})\|). \end{cases} \quad (7)$$

To satisfy this system, for the first equation in (7), it must hold that $\mathbf{n}_1^k = \mathbf{n}_2^k = \mathbf{0}$ due to the iteration-wise independence of the noise randomness. This implies $C_{1k} = C_{2k} = 0$ as:

$$\|\tilde{x} - \tilde{w}\| = 0, \quad \left\| \frac{1}{|D|} \sum_{i=1}^{|D|} \nabla f_i(\tilde{w}) \right\| = 0,$$

which forces $\tilde{x} = \tilde{w} = x^*$. Substituting this into (7), all conditions are satisfied. Therefore, it follows that a point is a fixed point of DP-C4 if and only if it is a first-order stationary point of the ERM problem (1), indicating DP-C4 eliminates solution bias.

In contrast, existing DP algorithms with constant clipping thresholds (e.g., DP-SGD, DP-SVRG, DiceSGD (Abadi et al., 2016; Lee, 2017; Zhang et al., 2023a)) do not admit fixed points, as the fixed-variance noise injected at each iteration continually disrupts equilibrium. For other schemes where clipping thresholds decay to 0, the persistent gradient estimation variance and gradual accumulation of clipping bias, combined with a mismatch between the decay rate of the thresholds and the convergence speed, lead to the fixed point being, with probability 1, not a solution to the original problem. This highlights the maximal advantage of DP-C4 in achieving solution debiasing. The detailed comparisons of different frameworks are provided in Appendix.B.

4. Theoretical Analysis

In this section, we investigate the privacy guarantees and convergence properties of DP-C4.

4.1. Privacy Analysis

Unlike prior DP frameworks, the data-dependent clipping thresholds in DP-C4 render standard sensitivity analyses inadequate. Specifically, existing analytical techniques either lead to overly pessimistic bounds or completely lose the

ability to bound the required noises in terms of the clipping thresholds, invalidating the discussion in Section.3.1.

To address this limitation, we develop a refined and extended sensitivity analysis. In particular, our analysis shows that, despite the apparent coupling between clipping thresholds and the data, the per-iteration noise can still be tightly calibrated in terms of C_{1k} and C_{2k} . Moreover, by exploiting the structural properties of DP-C4 and leveraging Rényi differential privacy (RDP) (Mironov, 2017), we obtain significantly reduced per-iteration noise bounds compared to standard DP-SGD-style analysis. These guarantees are formalized in the following theorem:

Theorem 1 (Noise Bounds). *Let $\theta = \frac{|D|^2}{|S|^2}$ and $\sigma^2 = \frac{4T(2 \log(1/\delta) + \epsilon)}{|S|^2 \epsilon^2}$. There exist σ_1^2, σ_2^2 defined in Alg.2 that guarantee (ϵ, δ) -DP of running DP-C4 with routine 1-4 for T iterations:*

$$\begin{cases} (\sigma_1^2, \sigma_2^2)_{R_{1 \& 2}} = \left((1 + \sqrt{\frac{p}{\theta}}) \sigma^2, \left(\frac{p}{\theta} + \sqrt{\frac{p}{\theta}} \right) \sigma^2 \right), \\ (\sigma_1^2, \sigma_2^2)_{R_{3 \& 4}} = \left((1 - p + \sqrt{\frac{p(1-p)}{\theta}}) \sigma^2, \left(\frac{p}{\theta} + \sqrt{\frac{p(1-p)}{\theta}} \right) \sigma^2 \right). \end{cases}$$

Proof sketch Our proof proceeds in three main steps:

1. We reformulate the DP-C4 workflow by explicitly separating the data dependence of the local variables $C_{(j)k}$ from that of the published iterates x^k and w^k (see Appendix.C.1).
2. We further generalize the definition of conditional sensitivity (see Def.5) and establish the associated DP composition result (see Prop.5), under which the variances of the Gaussian noises $\mathbf{n}_1^k$ and $\mathbf{n}_2^k$ admit tight upper bounds proportional to C_{1k} and C_{2k} .
3. By carefully examining the update rules of x^k and w^k , we allocate the privacy budget between g_1^k and g_2^k , thereby minimizing the per-iteration noise bound (see Paragraph.C.1).

Remark In practice, we choose the update probability $p = \frac{2|S|}{|D|} = \frac{2}{\sqrt{\theta}}$, guided by p is typically related to $\frac{|S|}{|D|}$ in SVRG (Kovalev et al., 2020). It follows directly that, $(\sigma_1^2 + \sigma_2^2)_{R_{1 \& 2}} = (1 + \sqrt{\frac{p}{\theta}})^2 \sigma^2 \approx \sigma^2$, $(\sigma_1^2 + \sigma_2^2)_{R_{3 \& 4}} = (\sqrt{\frac{p}{\theta}} + \sqrt{1-p})^2 \sigma^2 \approx (1 - O(p^3))^2 \sigma^2$ (see (23) and (25) in Appendix.C.1). At the k -th iteration, the upper bound of the total noise variance $C_{1k}^2 \sigma_1^2 + C_{2k}^2 \sigma_2^2$ is as follows:

$$\begin{aligned} (C_{1k}^2 \sigma_1^2 + C_{2k}^2 \sigma_2^2)_{R_{3 \& 4}} &\leq (\sigma_1^2 + \sigma_2^2) \max\{C_{1k}^2, C_{2k}^2\} \\ &\leq \sigma^2 (1 - O(p^3))^2 \max\{C_{1k}^2, C_{2k}^2\} < \sigma^2 \max\{C_{1k}^2, C_{2k}^2\} \\ &= \sigma^2 \min\{C^2, \max\{C_1^2 \|x^k - w^k\|^2, C_2^2 \|\nabla f(w^{k-1})\|^2\}\} \end{aligned}$$

It should be noted that σ^2 is exactly the noise multiplier in DP-SGD with a mini-batch size $|S|$. That is, for the same C , the total noise multiplier $(\sigma_1^2 + \sigma_2^2)$ in DP-C4 is approximately the same as that in DP-SGD, with noise variance

further decaying through C_{1k}^2 and C_{2k}^2 . The complete proof of Thm.1 is provided in Appendix.C. Appendix.C.2 further supplements our analysis by examining the sensitivity mismatch under the original thresholds in (5) and demonstrating the insufficiency of non-extended sensitivity analysis.

4.2. Convergence Analysis

We analyze the convergence properties of DP-C4 under two settings: (i) μ -strongly convex, and (ii) nonconvex. Our goal is to construct an unbiased Lyapunov function in strongly-convex case with specific clipping thresholds, and to establish convergence guarantees in non-convex case. It is worth emphasizing that these proofs are innovative in the following aspects: (1) existing DP algorithms lack solution-calibrated property and thus cannot employ unbiased Lyapunov functions for analysis; (2) by exploiting the unique structure of DP-C4, we carefully handle both noise variance and clipping bias, providing a novel perspective for the convergence analysis of DP optimization algorithms. We first present several assumptions:

Assumption 3.1 (Lower Bounded) $f(\cdot)$ is bounded from below by a finite constant f^* :

$$f(x) \geq f^* > -\infty, \forall x \in \mathbb{R}^d.$$

Assumption 3.2 (L-Smoothness) $f_i(\cdot)$ is L -smooth, i.e., it satisfies:

$$\|\nabla f_i(x) - \nabla f_i(y)\| \leq L\|x - y\|, \forall x, y \in \mathbb{R}^d.$$

Assumption 3.3 (μ -Strong Convexity) The loss function $f_i(\cdot)$ is μ -strongly convex:

$$f_i(y) \geq f_i(x) + \langle \nabla f_i(x), y - x \rangle + (\mu/2)\|x - y\|^2, \forall x, y \in \mathbb{R}^d.$$

Assumption 3.4 (Bounded Gradient Deviation) There exists a constant τ , such that:

$$\|\nabla f_i(x) - \nabla f(x)\| \leq \tau, \quad \forall i \in [N], \forall x \in \mathbb{R}^d.$$

Assumption 3.5 (Bounded Gradient). The gradient of the function is bounded in the sense that there exists a positive constant $G = \sup_{x \in \mathbb{R}^d, i \in [N]} \|\nabla f_i(x)\| < \infty$.

The above common assumptions serve as the foundation for analyzing DP algorithms (Abadi et al., 2016; Lee, 2017; Zhang et al., 2023a). We now turn to the convergence of DP-C4. To avoid overly intricate discussions, we restrict our setting to $C_{1k} = C_1\|x^k - w^k\|$, $C_{2k} = C_2\|\nabla f(w^{k-1})\|$. Let $\mathbb{E}[\cdot]$ and $\mathbb{E}_k[\cdot] := \mathbb{E}[\cdot|x^k, w^k]$ denote the full expectation and the conditional expectation based on the first k iterations of DP-C4, respectively. Then, we have:

Theorem 2 (Strongly Convex Case). Suppose Assumptions 3.1-3.5 hold. For any given $e > 0$ and constant DP noise multipliers σ_1, σ_2 , let $\{x^k\}_{k \geq 0}$ and $\{w^k\}_{k \geq 0}$ be generated

by Alg.2 with $\eta < \min\left\{\frac{\mu}{3N_1+A}, \frac{1}{2LN_2}\right\}$, $C_1 > 0$, $C_2 \geq \frac{\tau}{e} + 1$. When $\min\{\|\nabla f(w^k)\|, \|x^k - x^*\|, \|w^k - x^*\|\} > e$, define the Lyapunov function as:

$$\Phi^k := \mathbb{E}\|x^k - x^*\|^2 + \frac{2N_1\eta^2}{p}\mathbb{E}\|w^k - x^*\|^2 + \frac{2N_2\eta^2}{p}D^k,$$

where $D^k := \mathbb{E}\|\nabla f_i(w^k) - \nabla f_i(x^*)\|^2$, $N_1 := 8C_1^2(d\sigma_1^2 + 1) + \frac{4\eta^2}{pe^2}G^2C_2^2(d\sigma_2^2 + 1)$, $N_2 := 8C_2^2(d\sigma_2^2 + 1)$, $A := \frac{4G}{p\mu^2e^2}[(pC_2 + 1)L - C_1]\sqrt{2C_1^2(d\sigma_1^2 + 1) + \mu^2C_2^2(d\sigma_2^2 + 1)}$, and d denotes the model size. Then,

$$\Phi^{k+1} \leq \max\left\{1 - \mu\eta + (3N_1 + A)\eta^2, 1 - \frac{p}{2}\right\} \cdot \Phi^k < \Phi^k.$$

This indicates that DP-C4 converges to the optimum. In contrast to existing optimization algorithms whose convergence results typically rely on a single indicator, Thm.2 employs two accuracy indicators, Φ_k and $\min\{\|\nabla f(w^k)\|, \|x^k - x^*\|, \|w^k - x^*\|\}$. Specifically, for any given tolerance e , the Lyapunov function Φ^k decreases linearly until $\min\{\|\nabla f(w^k)\|, \|x^k - x^*\|, \|w^k - x^*\|\} \leq e$. In addition, achieving $\|x^k - x^*\| \leq e$ does not require choosing a large C_2 at the beginning of the algorithm. Instead, C_2 can be gradually increased during the convergence process to enforce convergence, thus avoiding the injection of excessive noise at the early stage. We now turn to the convergence analysis in the nonconvex setting:

Theorem 3 (Nonconvex Case). Suppose Assumptions 3.1, 3.2, 3.4, 3.5 hold. For any given constant DP noise multipliers σ_1, σ_2 and $C_1 \geq L$, $C_2 \geq 1$, let $\{x^k\}_{k=0}^T$ and $\{w^k\}_{k=0}^T$ be generated by Alg.2 with $\eta = \sqrt{\frac{2(f(x^0) - f(x^*))}{TLG(1+4G)}} = O(\frac{1}{\sqrt{T}})$. Then,

$$\begin{aligned} & \frac{1}{T} \sum_{k=1}^T \mathbb{E} \left[\lambda_1^k \|\nabla f(x^k)\|^2 + \lambda_2^k \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \right. \\ & \quad \left. + \lambda_3^k \|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \right] \\ & \leq 2 \sqrt{\frac{(f(x^0) - f(x^*))L\tilde{G}(1+4G)}{2T}} \\ & \quad + \frac{1}{T} \sum_{k=1}^T \mathbb{E} \left[\lambda_4^k \cdot 3\tau \|\nabla f(x^k)\| \right]. \end{aligned} \tag{8}$$

Here, $\tilde{G} := 4C^2(d\sigma_1^2 + 1) + 4G^2C_2^2(d\sigma_2^2 + 1)$, d denotes the

Table 1. Test accuracy (%) of different schemes on different datasets.

Method	SVM	CV Tasks			NLP Tasks	
	Mushroom	Mnist	Cifar-10	Cifar-100	IMDb	GLUE SST-2
DP-SGD	87.48	96.26	53.05	37.04	76.99	75.23
DP-SVRG	77.13	95.79	51.81	31.08	74.10	72.71
DiceSGD	90.65	97.02	60.24	40.73	78.19	78.71
DP-C4	96.98	97.16	64.50	43.12	81.23	82.24

model size, and for each k :

$$\begin{cases}
 \lambda_1^k := 1 - \frac{1}{3}(1 - \mathbb{P}_1^k)(2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}), \\
 \lambda_2^k := (1 - \mathbb{P}_2^k)(C_2 - 1), \\
 \lambda_3^k := (1 - \mathbb{P}_1^k)(C_1/L - 1), \\
 \lambda_4^k := \frac{1}{3}\mathbb{P}_1^k(2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}), \\
 \mathbb{P}_1^k := \Pr(\|\nabla f(x^k)\| \leq 3\tau \mid x^{k-1}), \\
 \mathbb{P}_1^k := \mathbb{E}_k[1_{\{\|\nabla f_i(x^k) - \nabla f_i(w^k)\| \leq C_{1k}\}}], \\
 \mathbb{P}_2^k := \mathbb{E}_k[1_{\{\|\nabla f_i(w^k)\| \leq C_{2k}\}}].
 \end{cases} \quad (9)$$

Since $\lambda_1^k \rightarrow 0$ and $\lambda_2^k \rightarrow 0$ require $\mathbb{P}_2^k \rightarrow 0$ and $\mathbb{P}_2^k \rightarrow 1$ simultaneously, λ_1^k and λ_2^k can not be zero synchronously. Thus, Thm.3 effectively characterizes convergence. It is worth noting that (8) is obtained by a piecewise discussion of $\|\nabla f(x^k)\|$ (more detail see Appendix.D.3): On the one hand, when $\|\nabla f(x^k)\| \geq 3\tau$, $\mathbb{E}\|\nabla f(x^k)\|^2$ exhibits strict descent (see (59) in Appendix.D.3), which guarantees that DP-C4 converges to the region $\|\nabla f(x^k)\| < 3\tau$. On the other hand, when $\|\nabla f(x^k)\| < 3\tau$, due to clipping bias, the right hand side introduces an optimization bias term $\frac{1}{T}\sum 3\tau\mathbb{E}[\lambda_4^k\|\nabla f(x^k)\|]$. Notably, it differs from previous work in the following aspects: (i) compared with a fixed clipping bias at the constant scale proportional to τ (Xiao et al., 2023), the optimization bias term in (8) is proportional to $\|\nabla f(x^k)\|$, which implies a gradually vanishing bias; (ii) at the k -th iteration, the last two terms on the left hand side of (8) also contribute to reducing the optimization bias. Employing the Cauchy-Schwarz inequality and setting $C_1 = L(C_2 - 1)\frac{1 - \mathbb{P}_2^k}{1 - \mathbb{P}_1^k} + L$, we obtain:

$$\begin{aligned}
 & 3\lambda_4^k\tau\|\nabla f(x^k)\| - \lambda_2^k\|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \\
 & - \lambda_3^k\|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \\
 & \leq 3\lambda_4^k\tau\|\nabla f(x^k)\| - \lambda_3^k\|\nabla f(x^k)\|^2 \\
 & \leq \frac{9(\lambda_4^k)^2\tau^2}{4(1 - \mathbb{P}_2^k)(C_2 - 1)} \xrightarrow{C_2 \rightarrow \infty} 0.
 \end{aligned}$$

This implies that by gradually increasing C_2 during iteration, together with the decaying step size, DP-C4 can converge to arbitrary precision.

5. Numerical Experiments

We conducted extensive experiments to demonstrate the advantages of DP-C4. Specifically, we evaluated our framework on Mushroom (mus, 1981), Mnist (Deng, 2012), Cifar-10, Cifar-100 (Krizhevsky et al., 2009), IMDb (Maas et al., 2011), and GLUE (Wang et al., 2018) datasets, comparing against both related baselines and state-of-the-art methods, namely DP-SGD (Abadi et al., 2016), DP-SVRG (Lee, 2017), and DiceSGD (Zhang et al., 2023a). In addition, we conducted a series of ablation studies on CIFAR-10 to systematically evaluate the effects of C_1 , C_2 , overall clipping threshold C , stepsize, different update routines, varying large-batch sizes, and update probabilities. Moreover, we present iteration-level statistics of C_{1k} and C_{2k} as empirical evidence of the effectiveness of the clipping strategy. The detailed results and discussions are provided in Appendix.E.

In the main experiments, we set the clipping thresholds to 1 for all methods, including C , C_1 , and C_2 in DP-C4. The step size η was tuned via grid search over $\{0.1, 0.05, 0.025, 0.0125\}$, and we report the best-performing results. For all mini-batches, we use a batch size of $|S| = 256$. In DP-C4 and DP-SVRG, we further set the large batch size to $|D'| = 4096$, and the update probability to $p = \frac{2|S|}{|D'|} = 0.125$. For the SVM task, we set the privacy parameters to $(\epsilon, \delta) = (1, 10^{-5})$, train for 50 epochs, and employ a logistic regression model on the Mushroom dataset. For image classification tasks, we set $(\epsilon, \delta) = (5, 10^{-5})$, train for 100 epochs, and adopt LeNet (LeCun et al., 2002) on Mnist, and ResNet20 (He et al., 2016) on CIFAR-10 and CIFAR-100. For NLP tasks, we set $(\epsilon, \delta) = (2, 10^{-5})$, train for 50 epochs, and adopt GRU-RNN (Cho et al., 2014) on both IMDb and GLUE. The results are summarized in Table.1, where we observe that DP-C4 consistently outperforms the baselines across SVM, CV, and NLP tasks.

6. Conclusion

In this work, we proposed DP-C4, which reconstructs the update rule and the clipping strategy of DP optimization to ensure that clipping bias and noise variance asymptotically vanish, thereby eliminating the solution bias inherent in ex-

isting frameworks. On the privacy side, we extended existing sensitivity analysis techniques and designed a structure-aware budget allocation tailored to the coupled clipping framework, leading to general (ϵ, δ) -DP guarantees. Compared with DP-SGD, our framework injects significantly less noise under the same privacy level. We established convergence guarantees by constructing an unbiased Lyapunov function under the μ -strongly convex setting and identifying a vanishing bias term in the general non-convex case, offering a novel perspective on DP optimization analysis. Experiments on SVM, CV, and NLP tasks demonstrate that DP-C4 consistently achieves superior privacy-utility trade-offs, underscoring its promise for practical deployment.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Property Details of DP-C4

This section provides a more detailed and formal discussion of two fundamental properties of DP-C4 that were stated in the main body of the paper. Specifically, we elaborate on the *consistently-vanishing error* property and the *solution-calibrated* property from a technical perspective.

Consistently-vanishing Error We point out that the clipping bias and noise variance of DP-C4 with substituted clipping thresholds also vanish. Specifically, continuing with the notation from Section 3.1, we have:

$$\begin{aligned} B_k + V_k &\leq \frac{|I_2^k|}{|S|^2} \sum_{i \in I_2^k} \left[\|\nabla f_S(x^k) - \nabla f_S(w^k)\| - C_1 \|x^k - w^k\| \right]^2 + \sigma_1^2 C_1^2 \|x^k - w^k\|^2 \\ &\quad + \left\| \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_2 \|\nabla f(w^{k-1})\|) - \nabla f(w^k) \right\|^2 + \sigma_2^2 C_2^2 \|\nabla f(w^{k-1})\|^2 \\ &\leq \left(\frac{|I_2^k|^2 (L - C_1)^2}{|S|^2} + \sigma_1^2 C_1^2 \right) \|x^k - w^k\|^2 + (\sigma_2^2 + 2) C_2^2 \|\nabla f(w^{k-1})\|^2 + 2 \|\nabla f(w^k)\|^2 \xrightarrow{x^k, w^k \rightarrow x^*} 0 \end{aligned}$$

At first glance, the above bound appears to rely on the L -smoothness assumption through the inequality $\|\nabla f(x^k) - \nabla f(w^k)\| \leq L \|x^k - w^k\|$. However, we emphasize that the vanishing behavior of $B_k + V_k$ does *not* fundamentally hinge on global L -smoothness. Instead, it follows from the *pointwise convergence* $x^k, w^k \rightarrow x^*$ together with the continuity of the gradient mapping. Specifically, since ∇f is continuous at x^* , we have

$$\|\nabla f(x^k) - \nabla f(w^k)\| = O(\|x^k - w^k\|), \quad \text{as } x^k, w^k \rightarrow x^*,$$

which implies that the coupled clipping bias vanishes even in the absence of a uniform Lipschitz constant. Similarly, the anchor-related terms vanish as $\nabla f(w^k) \rightarrow 0$ and $\nabla f(w^{k-1}) \rightarrow 0$, ensuring that both the clipping bias and the injected noise variance decay to zero along the optimization trajectory. Therefore, the asymptotic vanishing of $B_k + V_k$ is a consequence of local convergence and gradient continuity, rather than a property specific to L -smooth objectives.

Solution-Calibrated Property In this subsection, we formally leverage Proposition 4 to further substantiate the solution-calibrated property of DP-C4 stated in Section 3.4, and provide a rigorous proof. It is worth noting that DP-C4 involves multiple independent sources of randomness at each iteration, including subsampling, Gaussian noise injection, and randomized anchor updates. Therefore, fixed points must be characterized in an *almost-sure* sense.

Proposition 4 (Solution-Calibrated Property of DP-C4). *A pair $(\tilde{x}, \tilde{w})$ is a fixed point of DP-C4 with probability one if and only if*

$$\mathbb{P}(\tilde{x} = \tilde{w} = x^*) = 1,$$

where x^* is the first-order optimality condition of the ERM problem (1).

Proof. We prove both necessity and sufficiency.

(Necessity). Assume $(\tilde{x}, \tilde{w})$ is a fixed point of DP-C4 with probability one. By definition, this means that

$$\mathbb{P}(x^{k+1} = x^k = \tilde{x}, w^{k+1} = w^k = \tilde{w}) = 1.$$

From the update rule

$$x^{k+1} = x^k - \eta \tilde{g}^k,$$

we immediately obtain

$$\mathbb{P}(\tilde{g}^k = 0) = 1. \tag{10}$$

From Alg. 2, the noisy gradient estimator admits the decomposition

$$\tilde{g}^k = \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(\tilde{x}) - \nabla f_i(\tilde{w}), C_{1k}) + \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(\tilde{w}), C_{2k}) + \mathbf{n}_1^k + \mathbf{n}_2^k. \tag{11}$$

The noise terms satisfy

$$\mathbf{n}_1^k \sim \mathcal{N}(0, \sigma_1^2 C_{1k}^2 I), \quad \mathbf{n}_2^k \sim \mathcal{N}(0, \sigma_2^2 C_{2k}^2 I),$$

and are independent across iterations.

Since a non-degenerate Gaussian random vector equals zero with probability zero, the almost-sure equality in (10) implies

$$\mathbb{P}(C_{1k} = 0) = 1, \quad \mathbb{P}(C_{2k} = 0) = 1. \quad (12)$$

Recalling the definitions

$$C_{1k} = \min(C, C_1 \|\tilde{x} - \tilde{w}\|), \quad C_{2k} = \min(C, C_2 \|\nabla f(\tilde{w})\|),$$

we conclude that

$$\mathbb{P}(\|\tilde{x} - \tilde{w}\| = 0) = 1, \quad \mathbb{P}(\|\nabla f(\tilde{w})\| = 0) = 1.$$

Therefore,

$$\mathbb{P}(\tilde{x} = \tilde{w} = x^*) = 1.$$

(Sufficiency). Conversely, suppose

$$\mathbb{P}(\tilde{x} = \tilde{w} = x^*) = 1.$$

Then $\nabla f(\tilde{x}) = \nabla f(\tilde{w}) = 0$ almost surely, which directly implies

$$\mathbb{P}(C_{1k} = C_{2k} = 0) = 1.$$

As a result, all clipped gradient terms and injected noises vanish almost surely, yielding

$$\mathbb{P}(\tilde{g}^k = 0) = 1.$$

Hence,

$$\mathbb{P}(x^{k+1} = x^k = \tilde{x}, w^{k+1} = w^k = \tilde{w}) = 1,$$

which shows that $(\tilde{x} = x^*, \tilde{w} = x^*)$ is indeed a fixed point of DP-C4 with probability one. $\square$

B. Algorithm Comparison

In this section, we provide a detailed exposition of the fundamental distinction between DP-C4 and other algorithms (as an extension of Section 3.4), namely, the unique Solution-Calibrated Property that is exclusive to DP-C4 but absent in existing approaches.

In Section 3.4 and Appendix A, we have established the solution-calibrated property of DP-C4. In contrast, methods employing a constant clipping threshold (e.g., DP-SGD, DP-SVRG) do not admit fixed points, as the fixed-variance noise injected at each iteration continually disrupts equilibrium.

DP-SGD: For DP-SGD, suppose it admits a fixed point $\tilde{x}$, we obtain:

$$\tilde{x} = x^{k+1} = x^k - \eta \tilde{g}^k = \tilde{x} - \eta \tilde{g}^k,$$

That is, $\tilde{g}^k = \frac{1}{|S_k|} \sum_{i \in S_k} \text{clip}(\nabla f_i(\tilde{x}), C) + \mathbf{n}^k = 0$. However, due to the stochasticity introduced by the noise in each iteration, this condition cannot be satisfied with probability 1. Consequently, DP-SGD does not admit a fixed point.

DP-SVRG: The SVRG (Johnson & Zhang, 2013; Kovalev et al., 2020) method is a representative variance reduction technique. This method introduces an additional anchor point w^k , which is periodically updated and computed the full gradient. At k -th iteration, the gradient estimate is $g_{S_k}^k = \frac{1}{|S_k|} \sum_{i \in S_k} \nabla f_i(x^k) - \frac{1}{|S_k|} \sum_{i \in S_k} \nabla f_i(w^k) + \frac{1}{|D|} \sum_{i \in D} \nabla f_i(w^k)$, which is an unbiased estimate of the full gradient, i.e., $\mathbb{E}[g_{S_k}^k] = \nabla f(x^k)$. Moreover, it satisfies $g_{S_k}^k \xrightarrow{x^k, w^k \rightarrow x^*} \nabla f(x^*) = 0$. By integrating SVRG into the DP algorithm, DP-SVRG (Lee, 2017) has been proposed (see Alg.3). However, clipping $\nabla f_i(x^k)$ and $\nabla f_i(w^k)$ separately undermines the variance-reduction structure, where the resulting stochastic gradient becomes biased ($\mathbb{E} \tilde{g}^k \neq \nabla f(x^k)$) and no longer converges to zero as $x^k \rightarrow x^*$.

Algorithm 3 DP-SVRG (Lee, 2017)

```

1: Initialize  $x^0 = w^0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:   Sample  $S_k \subseteq D$ 
4:    $\tilde{g}_i^k(x) = \text{clip}(\nabla f_i(x^k), C) + \mathcal{N}(0, \sigma^2 C^2 I)$ 
5:    $\tilde{g}_i^k(w) = \text{clip}(\nabla f_i(w^k), C) + \mathcal{N}(0, \sigma^2 C^2 I)$ 
6:    $\tilde{g}^k = \frac{1}{|S_k|} \sum_{i \in S_k} \tilde{g}_i^k(x) - \frac{1}{|S_k|} \sum_{i \in S_k} \tilde{g}_i^k(w) + \frac{1}{|D|} \sum_{i \in D} \tilde{g}_i^k(w)$ 
7:    $x^{k+1} = x^k - \eta \tilde{g}^k$ 
8:    $w^{k+1} = \begin{cases} x^k, & \text{with probability } p \\ w^k, & \text{with probability } 1 - p \end{cases}$ 
9: end for
    
```

Decaying-Clipping-Threshold Schemes: For other schemes where the clipping threshold decays (TD) to 0, the persistent gradient estimation noise at each iteration, and the gradual accumulation of clipping bias, combined with a mismatch between the decay rate of the threshold and the convergence speed, ensures that the fixed point is, with probability 1, NOT a solution to the original problem. Taking DP-SGD^{TD} as an example, suppose it admits a fixed point $\tilde{x}$, we obtain:

$$\begin{cases} \tilde{g}^k = \frac{1}{|S_k|} \sum_{i \in S_k} \text{clip}(\nabla f_i(\tilde{x}), C_k) + \mathbf{n}^k = 0, \\ \mathbf{n}^k \sim \mathcal{N}(0, \sigma^2 C_k^2 I), \quad C_k \rightarrow 0 \end{cases}$$

We can observe that when the clipping threshold approaches zero (i.e., $C_k = 0$), the above equation is indeed satisfied, implying that DP-SGD with a decaying threshold admits a fixed point $\tilde{x}$. However, this fixed point arises from the elimination of the update due to the vanishing threshold, and therefore it does not guarantee that $\tilde{x} = x^*$.

Algorithm 4 DiceSGD (Zhang et al., 2023a)

```

1: Input: Dataset  $\mathcal{D}$ , learning rate  $\eta$ , clipping bounds  $C_1, C_2$ , noise scale  $\sigma$ , total steps  $T$ 
2: Output: Model parameters  $x^T$  satisfying  $(\epsilon, \delta)$ -DP
3: Initialize:  $e^0 = 0, x^0 \in \mathbb{R}^d$ 
4: for  $k = 0, \dots, T - 1$  do
5:   Randomly draw minibatch  $S$  from  $D$ 
6:    $g^k = \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k), C_1) + \text{clip}(e^k, C_2)$ 
7:    $x^{k+1} = x^k - \eta(g^k + \mathbf{n}^k)$ , where  $\mathbf{n}^k \sim \mathcal{N}(0, \sigma^2(C_1^2 + C_2^2)\mathbf{I})$ 
8:    $e^{k+1} = e^k + \frac{1}{|S|} \sum_{i \in S} \nabla f_i(x^k) - g^k$ 
9: end for
    
```

Error-Feedback-Embedded Schemes: By embedding error feedback (EF) into the iterative formulation, this class of methods alleviates the solution shift caused by differential privacy. In particular, the proposed DiceSGD (Zhang et al., 2023a) (Alg 4) eliminates the bias in each iteration *in expectation*. Therefore, in the sense of ignoring the injected noise and sampling randomness (i.e., in the full-expectation sense), it possesses a similar property. Assume that $(\tilde{x}, \tilde{e})$ is a fixed point of DiceSGD, then we have:

$$\begin{cases} \mathbb{E}[\tilde{x}] = \mathbb{E}[\tilde{x}] - \eta \mathbb{E}[g^k + \mathbf{n}^k] = \mathbb{E}[\tilde{x}] - \eta \mathbb{E}[g^k], \\ \mathbb{E}[\tilde{e}] = \mathbb{E}[\tilde{e}] + \mathbb{E}\left[\frac{1}{|S|} \sum_{i \in S} \nabla f_i(\mathbf{x}) - g^k\right] = \mathbb{E}[\tilde{e}] + \frac{1}{N} \sum_{i=1}^N \nabla f_i(\tilde{x}) - \mathbb{E}[g^k]. \end{cases} \quad (13)$$

We can verify that $(\tilde{x}, \tilde{e}) = (x^*, 0)$ is indeed a solution to (13), implying that, *in full-expectation sense*, the fixed point of DiceSGD coincides with the solution of the original problem. However, as discussed earlier, the randomness introduced by noise and sampling can disrupt this balance at any iteration, causing the iterates to deviate from the true solution.

Specifically, Table 2 summarizes the solution calibration property of different methods under both noise and sampling stochasticity, where the symbols $-$, $\checkmark$ and $\times$ respectively denote: no fixed point exists, the fixed point is (not) a solution to the problem. $\mathbb{E}_{\text{full}}$, $\mathbb{E}_{\text{noise}}$, $\mathbb{E}_{\text{sampling}}$, $\text{no-}\mathbb{E}$ denote, respectively, *in the sense of full expectation*, *in the sense of expectation over noise*, *in the sense of expectation over sampling*, and *taking into account all sources of randomness*.

Table 2. Algorithm Comparison on Solution-Calibrated Property.

Method Type of $\mathbb{E}$	DP-SGD ^(TD)	DP-SVRG ^(TD)	DiceSGD	DP-C4
$\mathbb{E}_{full}$	-($\times$)	-($\times$)	✓	✓
$\mathbb{E}_{noise}$	-($\times$)	-($\times$)	-	✓
$\mathbb{E}_{sampling}$	-($\times$)	-($\times$)	-	✓
no- $\mathbb{E}$	-($\times$)	-($\times$)	-	✓

C. Proofs of Privacy Analysis

In this section, we present the detailed proofs of the privacy results, i.e., Thm.1. We utilize Rényi Differential Privacy (RDP) as a bridge to analyze the privacy guarantees of DP-C4. Our insight is that each update of DP-C4 consists of two components, namely the *Coupled Term* and the *Anchor Term*, and we allocate different privacy budget weights to these components to discuss the corresponding noise levels. We first introduce several definitions and lemmas:

Definition 4 (Rényi Differential Privacy (RDP) (Mironov, 2017)). *A randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ satisfies (α, ε) -RDP ($\alpha \in (1, \infty)$, $\varepsilon > 0$) if for any datasets $D, D' \in \mathcal{D}$ with $d_H(D, D') = 1$, it holds that*

$$\frac{1}{\alpha - 1} \log \mathbb{E}_{o \sim \mathcal{M}(D')} \left[\left(\frac{\mathcal{M}(D)(o)}{\mathcal{M}(D')(o)} \right)^\alpha \right] \leq \varepsilon,$$

where $\mathcal{M}(D)(o)$ denotes the density of $\mathcal{M}(D)$ at o .

Lemma 2 (Post-processing Property of RDP (Mironov, 2017)). *Let $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ be (α, ε) -RDP and $g : \mathcal{R} \rightarrow \mathcal{R}'$ be any function. Then the composed mechanism $g \circ \mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}'$ is also (α, ε) -RDP.*

Lemma 3 (Composition of RDP Mechanisms (Mironov, 2017)). *Let $\mathcal{M}_r : \mathcal{R}_1 \times \dots \times \mathcal{R}_{r-1} \times \mathcal{D} \rightarrow \mathcal{R}_r$ be (α, ε_r) -RDP for $r \in [R]$. Then the mechanism*

$$\mathcal{M}(D) := (\mathcal{M}_1(D), \mathcal{M}_2(\mathcal{M}_1(D), D), \dots, \mathcal{M}_R(\mathcal{M}_1(D), \dots, D))$$

is $(\alpha, \sum_{r=1}^R \varepsilon_r)$ -RDP.

Lemma 4 (Conversion from RDP to DP (Mironov, 2017)). *If a mechanism $\mathcal{M}$ is (α, ε) -RDP, then $\mathcal{M}$ also satisfies $(\varepsilon + \frac{\log(1/\delta)}{\alpha-1}, \delta)$ -DP for any $\delta \in (0, 1)$.*

Lemma 5 (Gaussian Mechanism (Mironov, 2017)). *Given a function h , the Gaussian Mechanism*

$$\mathcal{M}(D) := h(D) + \mathcal{N}(0, \sigma^2 I)$$

satisfies $(\alpha, \alpha \Delta^2(h)/(2\sigma^2))$ -RDP for every $\alpha \in (1, \infty)$.

C.1. Proof of Theorem 1

We now outline the proof strategy of Theorem 1. The analysis proceeds in three conceptually distinct steps. First, we reformulate DP-C4 through an explicit separation between published variables and local (non-released) variables, and establish an *information decoupling* property across iterations. This enables us to recursively characterize the privacy loss of the released sequence via a conditional RDP analysis, rather than relying on a direct application of standard RDP composition. Second, leveraging this decoupled formulation, we introduce an extended notion of per-iteration conditional sensitivity, which captures the true sensitivity of the update mechanism when conditioned on the full history. This refinement allows us to precisely determine the noise magnitude required at each iteration. Finally, based on the resulting RDP bounds, we derive a fine-grained privacy budget allocation scheme by optimally distributing the total privacy budget across different gradient components, leading to tighter noise bounds and improved utility compared with existing baselines.

Information Decoupling in DP-C4 To derive a privacy guarantee for the proposed DP-C4 algorithm, we first reformulate the algorithm as the following procedures. In specific, we define $\mathcal{D}^k, \mathcal{X}^k, \mathcal{W}^k, \mathcal{C}_1^k, \mathcal{C}_2^k$ as the parameter spaces of dataset, the

iterates x^k, w^k and the clipping thresholds C_{1k}, C_{2k} at the k -th iteration, respectively:

$$\begin{aligned} \mathcal{A}^k &: \mathcal{D}^k \times \mathcal{X}^k \times \mathcal{W}^k \times \mathcal{C}_1^k \times \mathcal{C}_2^k \rightarrow \mathcal{X}^{k+1} \times \mathcal{W}^{k+1} \text{ as the evolution of } \mathbf{x}^{k+1} \text{ and } \mathbf{w}^{k+1}, \text{ i.e., lines 11,12 of Alg.2} \\ \mathcal{B}_1^k &: \mathcal{X}^{k+1} \times \mathcal{W}^{k+1} \rightarrow \mathcal{C}_1^{k+1} \text{ as the evolution of } C_{1(k+1)} \text{ at the } k\text{-th iteration, i.e., line 13 of Alg.2} \\ \mathcal{B}_2^k &: \mathcal{D}^k \times \mathcal{W}^k \rightarrow \mathcal{C}_2^{k+1} \text{ as the evolution of } C_{2(k+1)} \text{ at the } k\text{-th iteration, i.e., line 8 of Alg.2} \\ \mathcal{H}^k &: \prod_{t=0}^{k-1} \mathcal{D}^t \rightarrow \underbrace{\prod_{t=0}^k (\mathcal{X}^t \times \mathcal{W}^t)}_{\text{Published}:=\mathcal{P}^k} \times \underbrace{\prod_{t=0}^k (\mathcal{C}_1^t \times \mathcal{C}_2^t)}_{\text{Local}:=\mathcal{L}^k} \text{ as the evolution of all history } \{x^t, w^t\}_{t=0}^k, \{C_{1t}, C_{2t}\}_{t=0}^k \end{aligned}$$

Then we recursively bound the RDP guarantee of $\mathcal{H}^k$ for $k = 1, \dots, T$ using mathematical induction. We first provide the sketch of the proof as follows:

1. Show that the published sequence when $k = 1$, i.e., $\mathcal{P}^1 = \mathcal{H}^1/\mathcal{L}^1$ satisfies (α, ϵ^1) -RDP.
2. Assume that the output of $\mathcal{A}_1^t$ conditioned on the past information $\mathcal{H}^t$ satisfies (α, ϵ') -RDP. We derive the RDP guarantee for the published sequence at iteration $t + 1$, i.e., $\mathcal{P}^{t+1} = \mathcal{H}^{t+1}/\mathcal{L}^{t+1}$.

It is worth noting that this technique is fundamentally different from the proof strategies adopted in existing works as well as from approaches that directly rely on RDP composition (Lemma.3). On the one hand, we decouple and analyze, at a finer granularity, the respective impacts of released variables and local (non-released) variables on the output sequence at each iteration, which enables us to rigorously guarantee the privacy of DP-C4. On the other hand, observing that $\mathcal{P}^{t+1} = \mathcal{H}^{t+1}/\mathcal{L}^{t+1} = \{\mathcal{H}^t/\mathcal{L}^t, \mathcal{A}^t|\mathcal{H}^t\} = \{\mathcal{P}^t, \mathcal{A}^t|\mathcal{H}^t\}$, we are naturally led to a generalization of the classical notion of sensitivity (Def.5), which allows for a more accurate estimation of the noise magnitude required for privacy preservation (see the subsequent analysis).

Step 1: When $k = 1$, $\mathcal{P}^1 = \mathcal{H}^1/\mathcal{L}^1 = \mathcal{A}^1, \tilde{g}^0 = \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^0), C_2) + n_2^0$. Apply Lemma.5 and we obtain that $\mathcal{P}^1$ satisfies $(\alpha, \epsilon^1(\sigma_1, \sigma_2))$ -RDP where $\epsilon^1(\sigma_1, \sigma_2) \leq \frac{2\alpha}{|D|^2\sigma_2^2}$ is a function depending on the size of the injected noise σ .

Step 2:

Proposition 5 (Conditional Composition Mechanism). *Suppose $\mathcal{P}^t$ satisfies (α, ϵ) -RDP, $\mathcal{A}^t$ conditioned on $\mathcal{H}^t$ satisfies (α, ϵ') -RDP, then $\mathcal{P}^{t+1}$ satisfies $(\alpha, \epsilon + \epsilon')$ -RDP.*

Proof. Let us define $p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1}|\mathcal{H}^t)$ and $p'(\mathbf{x}^{t+1}, \mathbf{w}^{t+1}|\mathcal{H}^t)$ be the conditional probability-density-function (PDF) of the output of $\mathcal{A}^t$ on all history with neighboring datasets $\mathcal{D}$ and $\mathcal{D}'$, respectively; similarly, define $p(\mathcal{P}^t), p'(\mathcal{P}^t)$ be the PDF of the output of $\mathcal{P}^t$ with datasets $\mathcal{D}$ and $\mathcal{D}'$. Then we have:

$$p(\mathcal{P}^{t+1}) = p(\mathcal{P}^t, \mathbf{x}^{t+1}, \mathbf{w}^{t+1}) = p(\mathcal{P}^t)p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1}|\mathcal{P}^t) \stackrel{(a)}{=} p(\mathcal{P}^t)p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1}|\mathcal{P}^t, \mathcal{L}^t) = p(\mathcal{P}^t)p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1}|\mathcal{H}^t) \quad (14)$$

Here, step (a) follows from a conditional independence argument. Specifically, the local variables $\mathcal{L}^t := \{C_{1(0)}, C_{2(0)} \dots, C_{1t}, C_{2t}\}$ are generated according to $(C_{1t}, C_{2t}) = (\mathcal{B}_1^{t-1}(x^t, w^t), \mathcal{B}_2^{t-1}(w^{t-1}, D^{t-1}))$ and thus depend only on the published states (x^t, w^t) and past datasets D^{t-1} . On the other hand, the update $\mathbf{x}^{t+1}, \mathbf{w}^{t+1}$ is produced by $\mathcal{A}^t$ using the current dataset D^t together with (x^t, w^t) and C_{1t}, C_{2t} . Therefore, conditioning on the published history $\mathcal{P}^t = \{(x^s, w^s)\}_{s \leq t}$ already fixes all information contained in $\mathcal{L}^t$ that is relevant to the distribution of $\mathbf{x}^{t+1}$, and $\mathcal{L}^t$ introduces no additional randomness related to D^t . Formally, this implies the conditional independence $\mathbf{x}^{t+1} \perp\!\!\!\perp \mathcal{L}^t \mid \mathcal{P}^t$,

and hence $p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{P}^t) = p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{P}^t, \mathcal{L}^t) = p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{H}^t)$. Therefore, we have:

$$\begin{aligned}
 & \exp[(\alpha - 1)D_\alpha(\mathcal{P}^{t+1}(D) \parallel \mathcal{P}^{t+1}(D'))] \\
 &= \int_{\Pi_0^{t+1}(\mathcal{X} \times \mathcal{W})} p(\mathcal{P}^{t+1})^\alpha p'(\mathcal{P}^{t+1})^{1-\alpha} d\mathcal{P}^{t+1} \\
 &= \int_{\Pi_0^t(\mathcal{X} \times \mathcal{W})} \int_{\mathcal{X} \times \mathcal{W}} p(\mathcal{P}^t)^\alpha p'(\mathcal{P}^t)^{1-\alpha} p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{H}^t)^\alpha p'(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{H}^t)^{1-\alpha} d(\mathcal{P}^t, \mathbf{x}^{t+1}, \mathbf{w}^{t+1}) \\
 &= \int_{\Pi_0^t(\mathcal{X} \times \mathcal{W})} p(\mathcal{P}^t)^\alpha p'(\mathcal{P}^t)^{1-\alpha} d\mathcal{P}^t \int_{\mathcal{X} \times \mathcal{W}} p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{H}^t)^\alpha p'(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} \mid \mathcal{H}^t)^{1-\alpha} d(\mathbf{x}^{t+1}, \mathbf{w}^{t+1}) \\
 &\stackrel{(a)}{\leq} \exp((\alpha - 1)\epsilon) \exp((\alpha - 1)\epsilon') \\
 &= \exp((\alpha - 1)(\epsilon + \epsilon')),
 \end{aligned} \tag{15}$$

where (a) applies the assumptions that $\mathcal{P}^t$ satisfies (α, ϵ) -RDP and $\mathcal{A}^t \mid \mathcal{H}^t$ satisfies (α, ϵ') -RDP to the first and the second integration, respectively. Thus $\mathcal{P}^{t+1}$ satisfies $(\alpha, \epsilon + \epsilon')$ -RDP. $\square$

Extended Conditional Sensitivity Analysis It is worth noting that the above analysis reveals that, under the privacy guarantee of DP-C4, the amount of noise required at each iteration is determined by the conditional mechanism $\mathcal{A}^k \mid \mathcal{H}^k$, rather than by the unconditional mechanism $\mathcal{A}^k$ as in standard analyses. Motivated by this observation, we introduce a generalized notion of conditional sensitivity (Def.5) to precisely characterize the required noise magnitude. Accordingly, we obtain the following Lemma.6.

Definition 5 (Per-Iteration Conditional Sensitivity). *Let $\mathcal{M}(\cdot)$ be a randomized mechanism acting on a dataset D , and let $\mathcal{H}^k$ represent all the information (publicly revealed or local) history before the current iteration k . For any pair of neighboring datasets $D \sim D'$, the k^{th} -iteration conditional ℓ_2 -sensitivity of $\mathcal{M}$ is defined as*

$$\Delta_k(\mathcal{M}) \triangleq \sup_{D \sim D'} \|\mathcal{M}(D) - \mathcal{M}(D')\|_2 \mid \mathcal{H}^k.$$

Lemma 6 (k^{th} -iteration conditional ℓ_2 -sensitivity). *In Algorithm 2, the k^{th} -iteration conditional ℓ_2 -sensitivities of the Coupled Term g_1^k and the Anchor Term g_2^k are given by*

$$\Delta_{1k} = \frac{2C_{1k}}{|S|}, \quad \Delta_{2k} = \frac{2C_{2k}}{|D|}.$$

Proof. For the Coupled Term, we have:

$$g_1^k = \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}).$$

When $k \geq 1$, the k^{th} -iteration conditional ℓ_2 -sensitivity of g_1^k is bounded by

$$\begin{aligned}
 \max_{S, S'} \|g_1^k - g_1'^k\| \mid \mathcal{H}^k &= \max_{S, S'} \left\| \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}) - \frac{1}{|S'|} \sum_{i \in S'} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}) \right\| \mid C_{1k} \\
 &= \max_{S, S'} \frac{1}{|S|} \left\| \text{clip}(\nabla f_j(x^k) - \nabla f_j(w^k), C_{1k}) - \text{clip}(\nabla f'_j(x^k) - \nabla f'_j(w^k), C_{1k}) \right\| \mid C_{1k} \\
 &= \max_{S, S'} \frac{1}{|S|} \left\| \min \left\{ \frac{C_{1k}}{\|\nabla f_j(x^k) - \nabla f_j(w^k)\|}, 1 \right\} (\nabla f_j(x^k) - \nabla f_j(w^k)) \right. \\
 &\quad \left. - \min \left\{ \frac{C_{1k}}{\|\nabla f'_j(x^k) - \nabla f'_j(w^k)\|}, 1 \right\} (\nabla f'_j(x^k) - \nabla f'_j(w^k)) \right\| \mid C_{1k} \\
 &\leq \max_{S, S'} \frac{1}{|S|} \left(\left\| \min \left\{ \frac{C_{1k}}{\|\nabla f_j(x^k) - \nabla f_j(w^k)\|}, 1 \right\} (\nabla f_j(x^k) - \nabla f_j(w^k)) \right\| \right. \\
 &\quad \left. + \left\| \min \left\{ \frac{C_{1k}}{\|\nabla f'_j(x^k) - \nabla f'_j(w^k)\|}, 1 \right\} (\nabla f'_j(x^k) - \nabla f'_j(w^k)) \right\| \right) \mid C_{1k} \\
 &\leq \frac{2C_{1k}}{|S|} := \Delta_{1k}.
 \end{aligned}$$

For the Anchor Term, we have:

$$g_2^k = \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_{2k}).$$

When $k \geq 1$, the k^{th} -iteration conditional ℓ_2 -sensitivity of g_2^k can be bounded as

$$\begin{aligned} \max_{D, D'} \|g_2^k - g_2'^k\| \mid \mathcal{H}^k &= \max_{D, D'} \left\| \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_{2k}) - \frac{1}{|D'|} \sum_{i \in D'} \text{clip}(\nabla f_i(w^k), C_{2k}) \right\| \mid C_{2k} \\ &\leq \max_{D, D'} \frac{1}{|D|} \left\| \text{clip}(\nabla f_j(w^k), C_{2k}) - \text{clip}(\nabla f_j'(w^k), C_{2k}) \right\| \mid C_{2k} \\ &= \max_{D, D'} \frac{1}{|D|} \left\| \min \left\{ \frac{C_{2k}}{\|\nabla f_j(w^k)\|}, 1 \right\} \nabla f_j(w^k) - \min \left\{ \frac{C_{2k}}{\|\nabla f_j'(w^k)\|}, 1 \right\} \nabla f_j'(w^k) \right\| \mid C_{2k} \\ &\leq \max_{D, D'} \frac{1}{|D|} \left(\left\| \min \left\{ \frac{C_{2k}}{\|\nabla f_j(w^k)\|}, 1 \right\} \nabla f_j(w^k) \right\| + \left\| \min \left\{ \frac{C_{2k}}{\|\nabla f_j'(w^k)\|}, 1 \right\} \nabla f_j'(w^k) \right\| \right) \mid C_{2k} \\ &\leq \frac{2C_{2k}}{|D|} := \Delta_{2k}. \end{aligned}$$

Notably, when $k = 0$, the two terms also satisfy the aforementioned result, since $C_{1(0)}$ and $C_{2(0)}$ are constants by initialization. $\square$

Fine-Grained Privacy Budget Allocation With all the necessary preparations in place, we now proceed to the next step. We focus on analyzing Routines 1 and 2; the analysis for the remaining paths is similar, yielding similar conclusions. First, we derive an RDP bound for each term g_1^k and g_2^k .

For g_1^k , from Lemma.5, when we add noise $\mathbf{n}_1^k \sim \mathcal{N}(0, \sigma_1^2 C_{1k}^2)$, the term g_1^k satisfies $(\alpha, 2\alpha/(\sigma_1^2 \cdot |S|^2))$ -RDP, where the sensitivity of g_1^k is given in Lemma.6.

Similarly, for g_2^k , from Lemma.5, when we add noise $\mathbf{n}_2^k \sim \mathcal{N}(0, \sigma_2^2 C_{2k}^2)$, the term g_2^k satisfies $(\alpha, 2\alpha/(\sigma_2^2 \cdot |D|^2))$ -RDP, where the sensitivity of g_2^k is given in Lemma.6.

From Lemma.3, Alg.2 satisfies

$$\left(\alpha, \frac{2\alpha T}{\sigma_1^2 \cdot |S|^2} + \frac{2\alpha T p}{\sigma_2^2 \cdot |D|^2} \right)\text{-RDP}.$$

Then, by Lemma.4, it follows that Algorithm 2 satisfies

$$\left(\frac{2\alpha T}{\sigma_1^2 |S|^2} + \frac{2\alpha T p}{\sigma_2^2 |D|^2} + \frac{\log(1/\delta)}{\alpha - 1}, \delta \right)\text{-DP}.$$

For any target DP parameters $(\epsilon_{DP}, \delta_{DP})$, we discuss the variance of these noises through the allocation of the privacy budget. We set:

$$\begin{cases} \frac{1}{2}\epsilon_{DP} = \frac{\log(1/\delta)}{\alpha - 1} \\ \frac{1}{2}\epsilon_{DP} = \frac{2\alpha T}{\sigma_1^2 |S|^2} + \frac{2\alpha T p}{\sigma_2^2 |D|^2} \\ \delta_{DP} = \delta \end{cases} \quad (16)$$

From the first line of the above equation, we obtain $\alpha = 1 + 2\log(1/\delta_{DP})/\epsilon_{DP}$. In the following, under the constraint $\frac{1}{2}\epsilon_{DP} = \frac{2\alpha T}{\sigma_1^2 |S|^2} + \frac{2\alpha T p}{\sigma_2^2 |D|^2}$, we aim to minimize the total noise magnitude added to the gradient estimator per iteration, i.e., $\sigma_1^2 + \sigma_2^2$.

Let $\frac{2\alpha T}{\sigma_1^2 |S|^2} = \frac{1}{2}\beta\epsilon_{DP}$, $\frac{2\alpha T p}{\sigma_2^2 |D|^2} = \frac{1}{2}(1-\beta)\epsilon_{DP}$, where $\beta \in (0, 1)$. Solving for σ_1^2 and σ_2^2 yields:

$$\sigma_1^2 = \frac{4\alpha T}{\beta |S|^2 \epsilon_{DP}}, \quad \sigma_2^2 = \frac{4\alpha T p}{(1-\beta) |D|^2 \epsilon_{DP}} \quad (17)$$

Continuing the above objective, we aim to minimize the total noise per step by adjusting the budget allocation coefficient β , i.e., $\min_{\beta} \sigma_1^2 + \sigma_2^2$, and let $\theta = \frac{|D|^2}{|S|^2} \geq 1$. That is,

$$\min_{\beta \in (0, 1)} \frac{1}{\beta} + \frac{p}{(1-\beta)\theta} := y \quad (18)$$

Taking the derivative with respect to β and setting it to zero, we obtain:

$$\frac{dy}{d\beta} = -\frac{1}{\beta^2} + \frac{p}{\theta(1-\beta)^2} = 0 \quad (19)$$

Solving this, we obtain the value of β that minimizes $\min_{\beta} \sigma_1^2 + \sigma_2^2$ as:

$$\beta^* = \frac{1}{1 + \sqrt{\frac{p}{\theta}}} \quad (20)$$

Substituting back into (17), we obtain:

$$\begin{aligned} \sigma_1^2 &= \frac{4T(2\log(1/\delta_{DP}) + \epsilon_{DP})}{|S|^2\epsilon_{DP}^2} \cdot (1 + \sqrt{\frac{p}{\theta}}), \\ \sigma_2^2 &= \frac{4T(2\log(1/\delta_{DP}) + \epsilon_{DP})}{|D|^2\epsilon_{DP}^2} \cdot \sqrt{p} \cdot (\sqrt{\theta} + \sqrt{p}) \\ &= \frac{4T(2\log(1/\delta_{DP}) + \epsilon_{DP})}{|S|^2\epsilon_{DP}^2} \cdot (\frac{p}{\theta} + \sqrt{\frac{p}{\theta}}) \end{aligned} \quad (21)$$

Let $\sigma^2 = \frac{4T(2\log(1/\delta_{DP}) + \epsilon_{DP})}{|S|^2\epsilon_{DP}^2}$. It is straightforward to see that $\sigma^2 = \sigma_{\text{DP-SGD}}^2$ coincides exactly with the noise magnitude used in DP-SGD. In summary, we have:

$$(\sigma_1^2, \sigma_2^2)_{\text{Routine 1\&2}} = \left((1 + \sqrt{\frac{p}{\theta}})\sigma^2, (\frac{p}{\theta} + \sqrt{\frac{p}{\theta}})\sigma^2 \right) \quad (22)$$

For $(\sigma_1^2 + \sigma_2^2)_{\text{Routine 1\&2}}$, since $\frac{p}{\theta}$ is very small, we have:

$$\begin{aligned} (\sigma_1^2 + \sigma_2^2)_{\text{Routine 1\&2}} &= \left(1 + \sqrt{\frac{p}{\theta}} + \frac{p}{\theta} + \sqrt{\frac{p}{\theta}} \right) \sigma^2 \\ &= (1 + \sqrt{\frac{p}{\theta}})^2 \cdot \sigma^2 \approx \sigma^2 \end{aligned} \quad (23)$$

For Routines 3 and 4, since $g_1^k = 0$ when $w^{k+1} = x^{k+1}$, under T iterations, we only compute g_1^k for $T(1-p)$ rounds. Similarly, we can obtain:

$$(\sigma_1^2, \sigma_2^2)_{\text{Routine 3\&4}} = \left((1-p + \sqrt{\frac{p(1-p)}{\theta}})\sigma^2, (\frac{p}{\theta} + \sqrt{\frac{p(1-p)}{\theta}})\sigma^2 \right) \quad (24)$$

For $(\sigma_1^2 + \sigma_2^2)_{\text{Routine 3\&4}}$, let $p = \frac{2|S|}{|D|}$, we have:

$$\begin{aligned} (\sigma_1^2 + \sigma_2^2)_{\text{Routine 3\&4}} &= (1-p + \sqrt{\frac{p(1-p)}{\theta}} + \frac{p}{\theta} + \sqrt{\frac{p(1-p)}{\theta}})\sigma^2 \\ &= (\sqrt{1-p} + \sqrt{\frac{p}{\theta}})^2 \cdot \sigma^2 \\ &= (1 - \frac{p}{2} - \frac{p^2}{8} - O(p^3) + \frac{p^{\frac{3}{2}}}{2}) \cdot \sigma^2 \\ &= (1 - (\frac{p}{2} + \frac{p^2}{8}) - O(p^3) + \frac{p^{\frac{3}{2}}}{2}) \cdot \sigma^2 \\ &\leq (1 - O(p^3)) \cdot \sigma^2 < \sigma^2 \end{aligned} \quad (25)$$

For comparison, in the case of DP-SVRG, The noise added to the gradient estimator consists of three components. Similarly, we can derive that:

$$\begin{aligned} (\sigma_1^2 + \sigma_2^2 + \sigma_3^2)_{\text{Routine 1\&2}}^{\text{DP-SVRG}} &= (1 + \sqrt{\frac{2p}{\theta}})^2 \cdot \sigma^2 > \sigma^2 \\ (\sigma_1^2 + \sigma_2^2 + \sigma_3^2)_{\text{Routine 3\&4}}^{\text{DP-SVRG}} &= (\sqrt{2(1-p)} + \sqrt{\frac{p}{\theta}})^2 \cdot \sigma^2 > \sigma^2 \end{aligned} \quad (26)$$

From this, the multipliers for each kind of noise are summarized in the following table:

Table 3. Routine 1&2

Methods	σ_{DP-SGD}^2	$\sigma_{DP-SVRG}^2$	σ_{DP-C4}^2
Noise Multiplier	σ^2	$\sigma^2 \cdot (1 + \sqrt{\frac{2p}{\theta}})^2$	$\sigma^2 \cdot (1 + \sqrt{\frac{p}{\theta}})^2$
Comparison	$\sigma_{DP-SGD}^2 = \sigma^2$	$\sigma_{DP-SVRG}^2 > \sigma^2$	$\sigma_{DP-C4}^2 \approx \sigma^2$

Table 4. Routine 3&4

Methods	σ_{DP-SGD}^2	$\sigma_{DP-SVRG}^2$	σ_{DP-C4}^2
Noise Multiplier	σ^2	$\sigma^2 \cdot (\sqrt{2(1-p)} + \sqrt{\frac{p}{\theta}})^2$	$\sigma^2 \cdot (\sqrt{1-p} + \sqrt{\frac{p}{\theta}})^2$
Comparison	$\sigma_{DP-SGD}^2 = \sigma^2$	$\sigma_{DP-SVRG}^2 > \sigma^2$	$\sigma_{DP-C4}^2 < \sigma^2$

C.2. Necessity of Substituting C_{1k} and C_{2k}

In this subsection, we investigate from a privacy-analysis perspective why the substitution of C_{1k} and C_{2k} in Section.3.2 is necessary. We assert that under the original scheme design ((5), formally presented in Alg.5), namely

$$C'_{1k} = \min \left\{ C, C_1 \cdot \left\| \frac{1}{|S|} \sum_{i \in S} (\nabla f_i(x^k) - \nabla f_i(w^k)) \right\| \right\}, \quad C'_{2k} = \min \{ C, C_2 \cdot \|\nabla f(w^k)\| \},$$

a *sensitivity mismatch* inevitably arises. Specifically, the clipping thresholds depend on the data access of the current iteration, which breaks the alignment between the true conditional sensitivity of the released mechanism and the noise scale calibrated using these thresholds. As a consequence, the per-iteration sensitivity cannot be accurately characterized, rendering the corresponding privacy cost impossible to quantify precisely.

Similarly, we adopt the same notation and first reformulate Alg.5 as follows:

$$\begin{aligned} \mathcal{A}^k &: \mathcal{D}^k \times \mathcal{X}^k \times \mathcal{W}^k \times \mathcal{C}_1^k \times \mathcal{C}_2^k \rightarrow \mathcal{X}^{k+1} \times \mathcal{W}^{k+1} \text{ as the evolution of } \mathbf{x}^{k+1} \text{ and } \mathbf{w}^{k+1}, \text{ i.e., lines 12,13 of Alg.5} \\ \mathcal{B}_1^k &: \mathcal{D}^k \times \mathcal{X}^k \times \mathcal{W}^k \rightarrow \mathcal{C}_1^k \text{ as the evolution of } C'_{1k}, \text{ i.e., line 6 of Alg.5} \\ \mathcal{B}_2^k &: \mathcal{D}^k \times \mathcal{W}^k \rightarrow \mathcal{C}_2^k \text{ as the evolution of } C'_{2k}, \text{ i.e., line 7 of Alg.5} \\ \mathcal{H}^k &: \prod_{t=0}^{k-1} \mathcal{D}^t \rightarrow \underbrace{\prod_{t=0}^k (\mathcal{X}^t \times \mathcal{W}^t)}_{\text{Published} := \mathcal{P}^k} \times \underbrace{\prod_{t=0}^{k-1} (\mathcal{C}_1^t \times \mathcal{C}_2^t)}_{\text{Local} := \mathcal{L}^{k-1}} \text{ as the evolution of all history } \{x^t, w^t\}_{t=0}^k, \{C'_{1t}, C'_{2t}\}_{t=0}^{k-1} \end{aligned}$$

Notably, unlike DP-C4, Alg.5 does not employ the iterative-point-based and shifted clipping parameter estimation strategy. As a result, the determination of C'_{1k} and C'_{2k} directly relies on the dataset of the current iteration (see $\mathcal{B}_1^k, \mathcal{B}_2^k$), which causes a shift in the clipping threshold's history compared to DP-C4 (differences are highlighted). We have:

$$\begin{aligned} p(\mathcal{P}^{t+1}) &= p(\mathcal{P}^t, \mathbf{x}^{t+1}, \mathbf{w}^{t+1}) = p(\mathcal{P}^t) p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{P}^t) = p(\mathcal{P}^t) \underbrace{p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{P}^t, \mathcal{L}^{t-1})}_{= p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{H}^t)} \\ &\stackrel{(a)}{\neq} p(\mathcal{P}^t) p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{P}^t, \mathcal{L}^t) = p(\mathcal{P}^t) p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{H}^t, C'_{1t}, C'_{2t}), \end{aligned} \tag{27}$$

where (a) follows from the fact that $\mathcal{L}^t$ and $\mathbf{x}^{t+1} \mid \mathcal{P}^t$ are no longer conditionally independent, which implies $\frac{p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{P}^t)}{p(\mathbf{x}^{t+1}, \mathbf{w}^{t+1} | \mathcal{L}^t, \mathcal{P}^t)} \neq 1$. This indicates that C'_{1t} and C'_{2t} remain current-iteration-dependent even when conditioning

Algorithm 5 DP-C4'

```

1: Input: Dataset  $\mathcal{D}$ , learning rate  $\eta$ , clipping bounds  $C, C_1, C_2$ , noise scales  $\sigma_1, \sigma_2$ , total steps  $T$ , anchor update probability  $p$ 
2: Output: Model parameters  $x^T$  satisfying  $(\varepsilon, \delta)$ -DP
3: Initialize:  $x^0 = w^0 \in \mathbb{R}^d$ 
4: for  $k = 0$  to  $T - 1$  do
5:   Sample  $S \subseteq D$ 
6:    $C'_{1k} \leftarrow \min(C, C_1 \|\frac{1}{|S|} \sum_{i \in S} (\nabla f_i(x^k) - \nabla f_i(w^k))\|)$  {Coupled threshold}
7:    $C'_{2k} \leftarrow \min(C, C_2 \|\nabla f(w^k)\|)$  {Anchor threshold}
8:    $g_1^k \leftarrow \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C'_{1k})$  {Coupled term}
9:    $g_2^k \leftarrow \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C'_{2k})$  {Anchor term}
10:   $n_1^k \sim \mathcal{N}(0, \sigma_1^2 C_{1k}^2 I), \quad n_2^k \sim \mathcal{N}(0, \sigma_2^2 C_{2k}^2 I)$  {Sample DP noise}
11:   $\tilde{g}^k \leftarrow g_1^k + g_2^k + n_1^k + n_2^k$  {Add noise}
12:   $x^{k+1} \leftarrow x^k - \eta \cdot \tilde{g}^k$  {Update model}
13:   $w^{k+1} \leftarrow \begin{cases} x^k, & \text{with probability } p \\ w^k, & \text{with probability } 1 - p \end{cases}$  {Update anchor (Routine 1)}
14: end for
    
```

on $\mathcal{H}^t$ of Alg.5. In this case, C'_{1t} and C'_{2t} can no longer be treated as constants in the sensitivity analysis, and we obtain:

$$\begin{aligned}
 \Delta'_{1k} &= \max_{S, S'} \|g_1^k - g_1'^k\| \mid \mathcal{H}^k \\
 &= \max_{S, S'} \left\| \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}(S)) - \frac{1}{|S'|} \sum_{i \in S'} \text{clip}(\nabla f_i'(x^k) - \nabla f_i'(w^k), C_{1k}(S')) \right\| \\
 &\leq \max_{S, S'} \frac{1}{|S|} \sum_{i \in S} \left\| \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}(S)) - \text{clip}(\nabla f_i'(x^k) - \nabla f_i'(w^k), C_{1k}(S')) \right\| \\
 &\leq \max_{S, S'} \frac{1}{|S|} \sum_{i \in S} \left(\left\| \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}(S)) \right\| + \left\| \text{clip}(\nabla f_i'(x^k) - \nabla f_i'(w^k), C_{1k}(S')) \right\| \right) \\
 &\leq \max_{S, S'} (C'_{1k}(S) + C'_{1k}(S')) \\
 &\leq \max_{S, S'} \left(C'_{1k}(S) + \min(C, C_1 \|\frac{1}{|S'|} \sum_{i \in [S']} (\nabla f_i'(x^k) - \nabla f_i'(w^k))\|) \right) \\
 &\leq \max_{S, S'} \left(C'_{1k}(S) + \min(C, C_1 \|\frac{1}{|S'|} \sum_{i \in [S']} (\nabla f_i(x^k) - \nabla f_i(w^k)) - (\nabla f_j(x^k) - \nabla f_j(w^k)) + (\nabla f_j'(x^k) - \nabla f_j'(w^k))\|) \right) \\
 &\leq \max_{S, S'} \left(C'_{1k}(S) + \min(C, C_1 \|\frac{1}{|S'|} \sum_{i \in [S']} (\nabla f_i(x^k) - \nabla f_i(w^k))\| + \|(\nabla f_j(x^k) - \nabla f_j(w^k))\| + \|(\nabla f_j'(x^k) - \nabla f_j'(w^k))\|) \right) \\
 &\leq 2C'_{2k}(D) + \min(C, 4C_2G)
 \end{aligned} \tag{28}$$

$$\begin{aligned}
 \Delta'_{2k} &= \max_{D, D'} \|g_2^k - g_2'^k\| \mid \mathcal{H}^k \\
 &= \max_{D, D'} \left\| \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C'_{2k}(D)) - \frac{1}{|D'|} \sum_{i \in D'} \text{clip}(\nabla f'_i(w^k), C'_{2k}(D')) \right\| \\
 &\leq \max_{D, D'} \frac{1}{|D|} \sum_{i \in [|D|]} \|\text{clip}(\nabla f_i(w^k), C'_{2k}(D)) - \text{clip}(\nabla f'_i(w^k), C'_{2k}(D'))\| \\
 &\leq \max_{D, D'} \frac{1}{|D|} \sum_{i \in [|D|]} \left(\|\text{clip}(\nabla f_i(w^k), C'_{2k}(D))\| + \|\text{clip}(\nabla f'_i(w^k), C'_{2k}(D'))\| \right) \\
 &\leq \max_{D, D'} \left(C'_{2k}(D) + C'_{2k}(D') \right) \\
 &\leq \max_{D, D'} \left(C'_{2k}(D) + \min(C, C_2) \left\| \frac{1}{|D|} \left(\sum_{i \in [|D|]} \nabla f_i(w^k) - \nabla f_j(w^k) + \nabla f'_j(w^k) \right) \right\| \right) \\
 &\leq \max_{D, D'} \left(C'_{2k}(D) + \min(C, C_2) \left\| \frac{1}{|D|} \sum_{i \in [|D|]} \nabla f_i(w^k) \right\| + C_2 \|\nabla f_j(w^k)\| + C_2 \|\nabla f'_j(w^k)\| \right) \\
 &\leq 2C'_{2k}(D) + \min(C, 2C_2G)
 \end{aligned} \tag{29}$$

In this case, the sensitivity is no longer proportional to the clipping coefficients, which breaks the property described in Section 3.1 that the induced error vanishes asymptotically. We emphasize that conventional sensitivity analysis techniques cannot capture data dependencies across iterations, resulting in a pessimistic bound similar to (28) and (29) (even for Alg.2).

D. Proofs of Convergence Analysis

In this section, we present the detailed proofs of the convergence results of DP-C4, i.e., Lemma 1 and Thm.2-3.

D.1. Proof of Lemma 1

According to the definition of the clipping bias B^k (Section 3.1), we can directly obtain:

$$\begin{aligned}
 B^k &= \left\| \frac{1}{|S|} \sum_{i \in S} \text{clip}(h_i(x^k), C(\{h_i(x^k)\}_{i \in S})) - \frac{1}{|S|} \sum_{i \in S} h_i(x^k) \right\|_2^2 \\
 &\stackrel{(a)}{=} \left\| \frac{1}{|S|} \sum_{i \in I_1^k} h_i(x^k) + \frac{1}{|S|} \sum_{i \in I_2^k} \frac{C(\{h_i(x^k)\}_{i \in S})}{\|h_i(x^k)\|_2} \cdot h_i(x^k) \right. \\
 &\quad \left. - \frac{1}{|S|} \sum_{i \in I_1^k} h_i(x^k) - \frac{1}{|S|} \sum_{i \in I_2^k} h_i(x^k) \right\|_2^2 \\
 &= \left\| \frac{1}{|S|} \sum_{i \in I_2^k} \left(\frac{C(\{h_i(x^k)\}_{i \in S})}{\|h_i(x^k)\|_2} - 1 \right) \cdot h_i(x^k) \right\|_2^2 \\
 &= \frac{1}{|S|^2} \left\| \sum_{i \in I_2^k} \underbrace{(C(\{h_i(x^k)\}_{i \in S}) - \|h_i(x^k)\|_2)}_{\leq 0} \cdot \frac{h_i(x^k)}{\|h_i(x^k)\|} \right\|_2^2 \\
 &\leq \frac{1}{|S|^2} \left(\sum_{i \in I_2^k} (\|h_i(x^k)\|_2 - C(\{h_i(x^k)\}_{i \in S})) \cdot \frac{\|h_i(x^k)\|}{\|h_i(x^k)\|} \right)^2 \\
 &= \frac{1}{|S|^2} \left[\sum_{i \in I_2^k} (\|h_i(x^k)\|_2 - C(\{h_i(x^k)\}_{i \in S})) \right]^2 \\
 &\stackrel{(b)}{\leq} \frac{|I_2^k|}{|S|^2} \sum_{i \in I_2^k} \left[\|h_i(x^k)\|_2 - C(\{h_i(x^k)\}_{i \in S}) \right]^2
 \end{aligned} \tag{30}$$

D.2. Proof of Theorem 2

For the strongly convex case of DP-C4, our goal is to construct a Lyapunov function under appropriately chosen clipping coefficients. We first examine a potential term in the Lyapunov function of the system, namely $\mathbb{E}\|x^k - x^*\|^2$. Combining this with the update rule of DP-C4, and denoting the clipping biases as $b_1^k := \frac{1}{|S|} \sum_{i \in S} \text{clip}(\Delta_i^k, C_1 \|x^k - w^k\|) - \Delta_S^k$ and $b_2^k := \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_2 \|\nabla f(w^{k-1})\|) - \nabla f(w^k)$, where $\Delta_i^k := \nabla f_i(x^k) - \nabla f_i(w^k)$, $\Delta_S^k := \frac{1}{|S|} \sum_{i \in S} (\nabla f_i(x^k) - \nabla f_i(w^k))$. Then, we obtain:

$$\begin{aligned}
 \mathbb{E}_k \|x^{k+1} - x^*\|^2 &= \mathbb{E}_k \|x^k - x^* - \eta \tilde{g}^k\|^2 \\
 &= \|x^k - x^*\|^2 + \mathbb{E}_k [2\eta \langle \tilde{g}^k, x^* - x^k \rangle] + \eta^2 \mathbb{E}_k \|\tilde{g}^k\|^2 \\
 &\leq \|x^k - x^*\|^2 + 2\eta \mathbb{E}_k \langle \Delta_S^k + \nabla f(w^k) + b_1^k + b_2^k + n_1^k + n_2^k, x^* - x^k \rangle + \eta^2 \mathbb{E}_k \|\tilde{g}^k\|^2 \\
 &= \|x^k - x^*\|^2 + 2\eta \langle \nabla f(x^k), x^* - x^k \rangle + 2\eta \mathbb{E}_k \langle b_1^k + b_2^k + n_1^k + n_2^k, x^* - x^k \rangle + \eta^2 \mathbb{E}_k \|\tilde{g}^k\|^2 \\
 &\stackrel{(a)}{\leq} \|x^k - x^*\|^2 + 2\eta \underbrace{(f^* - f(x^k) - \frac{\mu}{2} \|x^k - x^*\|^2)}_{\mu\text{-strongly convex}} + 2\eta \mathbb{E}_k \langle b_1^k + b_2^k, x^* - x^k \rangle + \eta^2 \mathbb{E}_k \|\tilde{g}^k\|^2 \\
 &= \|x^k - x^*\|^2 (1 - \eta\mu) + 2\eta (f^* - f(x^k)) + 2\eta \mathbb{E}_k \langle b_1^k + b_2^k, x^* - x^k \rangle + \eta^2 \mathbb{E}_k \|\tilde{g}^k\|^2
 \end{aligned} \tag{31}$$

Here, (a) follows from the μ -strong convexity property, together with the fact that $\mathbb{E}_k[n_1^k] = \mathbb{E}_k[n_2^k] = 0$.

To bound the above terms, we now establish several auxiliary results. We first examine the randomness in the w^k iteration that arises both from the coin-flipping mechanism and from the stochasticity of the noise sampling. Owing to the independence of different sources of randomness, and in terms of the full expectation, we have:

$$\begin{aligned}
 &\mathbb{E}[\|x^k - w^k\| \cdot \|x^k - x^*\|] \\
 &\stackrel{(a)}{\leq} \mathbb{E}[\|x^k - w^k\| \cdot \frac{\|\nabla f(x^k) - \nabla f(x^*)\|}{\mu}] \\
 &= \frac{1}{\mu} \mathbb{E}[\|x^k - w^k\| \cdot \|\nabla f(x^k)\|] \\
 &\stackrel{(b)}{\leq} \frac{G}{\mu} \mathbb{E}\|x^k - w^k\| \\
 &\stackrel{(c)}{=} \frac{G}{\mu} \mathbb{E}[p\|x^k - x^{k-1}\| + (1-p)\|x^k - w^{k-1}\|] \\
 &\stackrel{(d)}{\leq} \frac{G}{\mu} \mathbb{E}[\|x^k - x^{k-1}\| + (1-p)\|x^{k-1} - w^{k-1}\|] \\
 &\leq \frac{G}{\mu} \mathbb{E}[(\|x^k - x^{k-1}\| + (1-p)\|x^{k-1} - x^{k-2}\| + (1-p)^2\|x^{k-2} - x^{k-3}\| + \dots \\
 &\quad + (1-p)^{k-1}\|x^1 - x^0\| + p\|w^0 - x^0\|] \\
 &\leq \frac{G}{\mu} \mathbb{E}[\eta(\|\tilde{g}^{k-1}\| + (1-p)\|\tilde{g}^{k-2}\| + \dots + (1-p)^k\|\tilde{g}^0\|)] \\
 &\leq \frac{G}{\mu} \mathbb{E}[\eta \tilde{G}(1 + (1-p) + (1-p)^2 + \dots + (1-p)^k)] \\
 &\leq \eta \frac{G\tilde{G}}{p\mu e^2} \cdot e^2 \leq \eta \frac{G\tilde{G}}{p\mu e^2} \mathbb{E}\|x^k - x^*\|^2
 \end{aligned} \tag{32}$$

Here, (a) follows from the μ -strong convexity property; (b) is due to Assumption 3.5; (c) comes from the iterative update rule of w^k ; and (d) is obtained by applying the triangle inequality, $\tilde{G}$ is given by (35). Similarly, we also obtain the following

results, which will be used in the subsequent proofs:

$$\begin{aligned}
 \mathbb{E} \|x^{k-1} - w^{k-1}\| \cdot \|x^k - x^\star\| &\leq \eta \frac{G\tilde{G}}{p\mu} \\
 \mathbb{E} \|w^k - w^{k-1}\|^2 &\leq \eta^2 \frac{\tilde{G}^2}{p} \\
 \mathbb{E} \|x^{k-1} - w^{k-1}\| \cdot \|\nabla f(x^k)\| &\leq \eta \frac{G\tilde{G}}{p}
 \end{aligned} \tag{33}$$

Next, we derive upper bounds for the last two terms in the above expression. Specifically, we begin by analyzing the upper bound of $\mathbb{E}_k \|\tilde{g}^k\|^2$. For any precision $e > 0$, when $\|\nabla f(w)\|, \|x^k - x^\star\|, \|w^k - x^\star\| > e$, we have:

$$\begin{aligned}
 \mathbb{E} \|\tilde{g}^k\|^2 &= \mathbb{E} \left[\left\| \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_1 \|x^k - w^k\|) \right\|^2 \right. \\
 &\quad \left. + \frac{1}{N} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_2 \|\nabla f(w^{k-1})\|) + n_1^k + n_2^k \right]^2 \\
 &\stackrel{(a)}{\leq} 4\mathbb{E} \left\| \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_1 \|x^k - w^k\|) \right\|^2 \\
 &\quad + 4\mathbb{E} \left\| \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_2 \|\nabla f(w^{k-1})\|) \right\|^2 \\
 &\quad + 4d\sigma_1^2 C_1^2 \mathbb{E} \|x^k - w^k\|^2 + 4d\sigma_2^2 C_2^2 \mathbb{E} \|\nabla f(w^{k-1})\|^2 \\
 &\leq 4C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|x^k - x^\star + x^\star - w^k\|^2 \\
 &\quad + 4C_2^2 (d\sigma_2^2 + 1) \mathbb{E} \|\nabla f(w^k) - (\nabla f(w^k) - \nabla f(w^{k-1}))\|^2 \\
 &\leq 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|x^k - x^\star\|^2 + 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|w^k - x^\star\|^2 \\
 &\quad + 8C_2^2 (d\sigma_2^2 + 1) \mathbb{E} \|\nabla f(w^k)\|^2 + 8C_2^2 (d\sigma_2^2 + 1) \mathbb{E} \|\nabla f(w^k) - \nabla f(w^{k-1})\|^2 \\
 &\stackrel{(b)}{\leq} 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|x^k - x^\star\|^2 + 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|w^k - x^\star\|^2 \\
 &\quad + 8C_2^2 (d\sigma_2^2 + 1) \mathbb{E} \|\nabla f(w^k)\|^2 + 8C_2^2 L^2 (d\sigma_2^2 + 1) \mathbb{E} \|w^k - w^{k-1}\|^2 \\
 &\stackrel{(c)}{\leq} 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|x^k - x^\star\|^2 + 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|w^k - x^\star\|^2 \\
 &\quad + 8C_2^2 (d\sigma_2^2 + 1) \mathbb{E} \|\nabla f(w^k)\|^2 + 8C_2^2 L^2 (d\sigma_2^2 + 1) \eta^2 \frac{\tilde{G}^2}{pe^2} \cdot e^2 \\
 &\leq 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|x^k - x^\star\|^2 + 8C_1^2 (d\sigma_1^2 + 1) \mathbb{E} \|w^k - x^\star\|^2 \\
 &\quad + 8C_2^2 (d\sigma_2^2 + 1) \mathbb{E} \|\nabla f(w^k)\|^2 + 4C_2^2 L^2 (d\sigma_2^2 + 1) \eta^2 \frac{\tilde{G}^2}{pe^2} (\mathbb{E} \|x^k - x^\star\|^2 + \mathbb{E} \|w^k - x^\star\|^2) \\
 &\stackrel{(d)}{\leq} \underbrace{(8C_1^2 (d\sigma_1^2 + 1) + \frac{4\eta^2}{pe^2} G^2 C_2^2 (d\sigma_2^2 + 1))}_{:=N_1} (\mathbb{E} \|x^k - x^\star\|^2 + \mathbb{E} \|w^k - x^\star\|^2) \\
 &\quad + \underbrace{8C_2^2 (d\sigma_2^2 + 1)}_{:=N_2} \underbrace{\mathbb{E} \|\nabla f_i(w^k) - \nabla f_i(x^\star)\|^2}_{:=D^k}.
 \end{aligned} \tag{34}$$

Where d denotes the model size. (a) follows from the Cauchy-Schwarz inequality; (b) follows from the L -smooth property; (c) follows from (33), $\tilde{G}$ is given by (35); and (d) uses the Cauchy-Schwarz inequality along with the convexity of the squared ℓ_2 -norm, i.e., $\|\mathbb{E}[X]\|^2 \leq \mathbb{E}[\|X\|^2]$. For $\mathbb{E} \|\tilde{g}^k\|$, we have:

$$\begin{aligned}
 \mathbb{E} \|\tilde{g}^k\| &\leq \sqrt{\mathbb{E} \|\tilde{g}^k\|^2} \\
 &\leq (4C_1^2(d\sigma_1^2+1)\|x^k-w^k\|^2+4C_2^2(d\sigma_2^2+1)\|\nabla f(w^{k-1})\|^2)^{\frac{1}{2}} \\
 &\leq \left(\frac{4C_1^2(d\sigma_1^2+1)}{\mu^2}\|\nabla f(x^k)-\nabla f(w^k)\|^2+4C_2^2(d\sigma_2^2+1)\|\nabla f(w^{k-1})\|^2\right)^{\frac{1}{2}} \\
 &\leq \left(\frac{8C_1^2(d\sigma_1^2+1)G^2}{\mu^2}+4C_2^2(d\sigma_2^2+1)G^2\right)^{\frac{1}{2}} \\
 &= \frac{2G}{\mu} \sqrt{2C_1^2(d\sigma_1^2+1)+\mu^2C_2^2(d\sigma_2^2+1)} := \tilde{G}
 \end{aligned} \tag{35}$$

We define the unclipped and clipped sample sets for the first clipping as $J_1^k := \{j : \|\Delta_j^k\| \leq C_{1k}\}$ and $J_2^k := \{j : \|\Delta_j^k\| > C_{1k}\}$, and those induced by the second clipping as $I_1^k := \{i : \|\nabla f_i(w^k)\| \leq C_{2k}\}$ and $I_2^k := \{i : \|\nabla f_i(w^k)\| > C_{2k}\}$. By choosing $C_1 > 0$ and $C_2 \geq \frac{\tau}{\epsilon} + 1$, we have:

$$\begin{aligned}
 b_1^k &= \frac{1}{|S|} \sum_{i \in S} \text{clip}(\Delta_i^k, C_{1k}) - \Delta_S^k = \frac{1}{|S|} \left(\sum_{i \in J_1^k} \Delta_i^k + \sum_{i \in J_2^k} \frac{C_{1k}}{\|\Delta_i^k\|} \Delta_i^k \right) - \Delta_S^k \\
 &= \frac{1}{|S|} \sum_{i \in J_2^k} \left(\frac{C_{1k}}{\|\Delta_i^k\|} - 1 \right) \cdot \Delta_i^k = \frac{1}{|S|} \sum_{i \in J_2^k} \underbrace{(C_1\|x^k-w^k\| - \|\Delta_i^k\|)}_{<0} \cdot \frac{\Delta_i^k}{\|\Delta_i^k\|}
 \end{aligned} \tag{36}$$

$$\begin{aligned}
 b_2^k &= \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_{2k}) - \nabla f(w^k) = \frac{1}{|D|} \left(\sum_{i \in I_1^k} \nabla f_i(w^k) + \sum_{i \in I_2^k} \frac{C_{2k}}{\|\nabla f_i(w^k)\|} \nabla f_i(w^k) \right) - \nabla f(w^k) \\
 &= \frac{1}{|D|} \sum_{i \in I_2^k} \left(\frac{C_{2k}}{\|\nabla f_i(w^k)\|} - 1 \right) \cdot \nabla f_i(w^k) = \frac{1}{|D|} \sum_{i \in I_2^k} \underbrace{(C_2\|\nabla f(w^{k-1})\| - \|\nabla f_i(w^k)\|)}_{<0} \cdot \frac{\nabla f_i(w^k)}{\|\nabla f_i(w^k)\|}
 \end{aligned} \tag{37}$$

For the first clipping, we define the probability of an individual sample remaining unclipped as $\mathbb{P}_1^k := \mathbb{E}_k 1_{\{\|\Delta_i^k\| \leq C_{1k}\}}$. Then, we have:

$$\begin{aligned}
 \mathbb{E}_k[\langle b_1^k, x^* - x^k \rangle] &= \mathbb{E}_k \left[\left\langle \frac{1}{|S|} \sum_{i \in J_2^k} \underbrace{(C_1\|x^k-w^k\| - \|\Delta_i^k\|)}_{<0} \cdot \frac{\Delta_i^k}{\|\Delta_i^k\|}, x^* - x^k \right\rangle \right] \\
 &\stackrel{(a)}{\leq} \mathbb{E}_k \left[\frac{1}{|S|} \sum_{i \in J_2^k} (\|\Delta_i^k\| - C_1\|x^k-w^k\|) \cdot \|x^k - x^*\| \right] \\
 &\stackrel{(b)}{\leq} \mathbb{E}_k \left[\frac{1}{|S|} \sum_{i \in J_2^k} (L\|x^k-w^k\| - C_1\|x^k-w^k\|) \cdot \|x^k - x^*\| \right] \\
 &\leq \mathbb{E}_k(1 - \mathbb{P}_1^k)(L - C_1)\|x^k - w^k\| \cdot \|x^k - x^*\| \\
 &\stackrel{(c)}{\leq} \eta(L - C_1) \frac{G\tilde{G}}{\mu\epsilon^2} \mathbb{E}_k \|x^k - x^*\|^2
 \end{aligned} \tag{38}$$

Here, (a) follows from the Cauchy–Schwarz inequality; (b) follows from the L -smoothness of the objective function; (c)

follows directly from (32). For the second clipping, similarly, we define $\mathbb{P}_2^k := \mathbb{E}_k 1_{\{\|\nabla f_i(w^k)\| \leq C_{2k}\}}$, we have:

$$\begin{aligned}
 \mathbb{E}_k \langle b_2^k, x^* - x^k \rangle &= \mathbb{E}_k \left\langle \frac{1}{|D|} \sum_{i \in J_2^k} \underbrace{(C_2 \|\nabla f(w^{k-1})\| - \|\nabla f_i(w^k)\|)}_{<0} \frac{\nabla f_i(w^k)}{\|\nabla f_i(w^k)\|}, x^* - x^k \right\rangle \\
 &\stackrel{(a)}{\leq} \mathbb{E}_k \left| \frac{1}{|D|} \sum_{i \in J_2^k} (\|\nabla f_i(w^k)\| - C_2 \|\nabla f(w^{k-1})\|) \cdot \|x^k - x^*\| \right| \\
 &\stackrel{(b)}{\leq} \mathbb{E}_k \left| \frac{1}{|D|} \sum_{i \in J_2^k} (\|\nabla f_i(w^k)\| - C_2 \|\nabla f(w^k)\| + C_2 \|\nabla f(w^k) - \nabla f(w^{k-1})\|) \cdot \|x^k - x^*\| \right| \\
 &\stackrel{(c)}{\leq} \mathbb{E}_k \left| \frac{1}{|D|} \sum_{i \in J_2^k} \underbrace{((\tau - (C_2 - 1) \|\nabla f(w^k)\|) + C_2 \|\nabla f(w^k) - \nabla f(w^{k-1})\|)}_{\leq 0} \cdot \|x^k - x^*\| \right| \\
 &\leq \mathbb{E}_k \left| \frac{1}{|D|} \sum_{i \in J_2^k} C_2 \|\nabla f(w^k) - \nabla f(w^{k-1})\| \cdot \|x^k - x^*\| \right| \\
 &\leq \mathbb{E}_k (1 - \mathbb{P}_2^k) L C_2 \|w^k - w^{k-1}\| \cdot \|x^k - x^*\| \\
 &\stackrel{(d)}{\leq} L p C_2 \mathbb{E}_k \|x^{k-1} - w^{k-1}\| \cdot \|x^k - x^*\| \\
 &\stackrel{(e)}{\leq} \eta L p C_2 \frac{G \tilde{G}}{p \mu e^2} \cdot e^2 \\
 &\stackrel{(f)}{\leq} \eta L C_2 \frac{G \tilde{G}}{\mu e^2} \cdot \mathbb{E} \|x^k - x^*\|^2
 \end{aligned} \tag{39}$$

Here, (a) follows from the Cauchy–Schwarz inequality; (b) from the triangle inequality; (c) from Assumption 3.4 and the choice of C_2 ; (d) comes from the iterative update rule of w^k ; (e) follows directly from (33); and (f) is due to the precision conditions we imposed.

With these preparations in place, we are now ready to proceed. For notational simplicity, in (34) we define $D^k := \mathbb{E} \|\nabla f_i(w^k) - \nabla f_i(x^*)\|^2$, $N_1 := 8C_1^2(d\sigma_1^2+1) + \frac{4\eta^2}{p\epsilon^2} G^2 C_2^2(d\sigma_2^2+1)$, $N_2 := 8C_2^2(d\sigma_2^2+1)$, and $A := \frac{2G\tilde{G}}{p\mu e^2} [(pC_2+1)L - C_1]$. Substituting (32)–(39) into (31), and taking the full expectation on both sides of (31), (34), (38) and (39), we obtain:

$$\begin{aligned}
 \mathbb{E} \|x^{k+1} - x^*\|^2 &\leq (1 - \eta\mu + \eta^2(N_1 + A)) \mathbb{E} \|x^k - x^*\|^2 + \eta^2 N_1 \mathbb{E} \|w^k - x^*\|^2 \\
 &\quad + \eta^2 N_2 D^k - 2\eta(\mathbb{E} f(x^k) - f^*)
 \end{aligned} \tag{40}$$

We now consider the iterative update of $\{w^k\}_{k \in [T]}$ in DP-C4 (Line 5 in Alg.2). Since w^k is updated with a certain probability, we have:

$$\begin{aligned}
 \mathbb{E} \|w^{k+1} - x^*\|^2 &= p \mathbb{E} \|x^k - x^*\|^2 + (1 - p) \mathbb{E} \|w^k - x^*\|^2 \\
 D^{k+1} &= (1 - p) D^k + p \mathbb{E} \|\nabla f_i(x^k) - \nabla f_i(x^*)\|^2 \\
 &\leq (1 - p) D^k + 2Lp(\mathbb{E} f(x^k) - f^*)
 \end{aligned} \tag{41}$$

We define the Lyapunov function of DP-C4 as follows:

$$\begin{aligned}
 \Phi^k &= \mathbb{E} \|x^k - x^*\|^2 + \frac{2N_1\eta^2}{p} \mathbb{E} \|w^k - x^*\|^2 + \frac{2N_2\eta^2}{p} D^k \\
 &= \mathbb{E} \|x^k - x^*\|^2 + \frac{16L^2C_1^2(d\sigma_1^2+1)\eta^2}{p} \mathbb{E} \|w^k - x^*\|^2 + \frac{8C_2^2(d\sigma_2^2+1)\eta^2}{p} D^k
 \end{aligned} \tag{42}$$

Let $\eta < \min \left\{ \frac{\mu}{3N_1+A}, \frac{1}{2LN_2} \right\}$. Then, we observe that:

$$\begin{aligned}
 \Phi^{k+1} &= \mathbb{E} \|x^{k+1} - x^*\|^2 + \frac{2N_1\eta^2}{p} \mathbb{E} \|w^{k+1} - x^*\|^2 + \frac{2N_2\eta^2}{p} D^{k+1} \\
 &\leq (1 - \mu\eta + (N_1 + A)\eta^2 + p \frac{2N_1\eta^2}{p}) \mathbb{E} \|x^k - x^*\|^2 + (N_1\eta^2 + (1-p) \frac{2N_1\eta^2}{p}) \mathbb{E} \|w^k - x^*\|^2 \\
 &\quad + (N_2\eta^2 + (1-p) \frac{2N_2\eta^2}{p}) D^k + \underbrace{(4LN_2\eta^2 - 2\eta)}_{<0} (\mathbb{E} f(x^k) - f^*) \\
 &= \underbrace{(1 - \mu\eta + (3N_1 + A)\eta^2)}_{<1} \mathbb{E} \|x^k - x^*\|^2 + (1 - \frac{p}{2}) \frac{2N_1\eta^2}{p} \mathbb{E} \|w^k - x^*\|^2 + (1 - \frac{p}{2}) \frac{2N_2\eta^2}{p} D^k
 \end{aligned} \tag{43}$$

That is,

$$\Phi^{k+1} \leq \underbrace{\max\{1 - \mu\eta + (3N_1 + A)\eta^2, 1 - \frac{p}{2}\}}_{<1} \cdot \Phi^k < \Phi^k, \tag{44}$$

which implies an exponential decay of the Lyapunov function.

D.3. Proof of Theorem 3

Unlike Thm.2, here we study the general convergence analysis in the non-convex setting without imposing accuracy-dependent restrictions on the clipping coefficients C_1 and C_2 . Therefore, we need to consider the clipping bias in a more refined manner. First, since $f(x)$ is L -smooth, we have:

$$\begin{aligned}
 f(x^{k+1}) - f(x^k) &\leq \langle \nabla f(x^k), x^{k+1} - x^k \rangle + \frac{L}{2} \|x^{k+1} - x^k\|^2 \\
 &= -\eta \langle \nabla f(x^k), \tilde{g}^k \rangle + \frac{L\eta^2}{2} \|\tilde{g}^k\|^2.
 \end{aligned} \tag{45}$$

Taking the expectation on both sides of the inequality, and let us define $g^k := \tilde{g}^k - n_1^k - n_2^k$, we obtain:

$$\mathbb{E}_k[f(x^{k+1})] - f(x^k) \leq -\eta \mathbb{E}_k \langle \nabla f(x^k), g^k \rangle + \frac{L\eta^2}{2} \mathbb{E}_k[\|g^k + n_1^k + n_2^k\|^2] \tag{46}$$

Our current goal is to derive a lower bound for $\mathbb{E}_k \langle \nabla f(x^k), g^k \rangle$ and an upper bound for $\mathbb{E}_k[\|g^k + n_1^k + n_2^k\|^2]$. We first consider the upper bound of $\mathbb{E}_k[\|g^k + n_1^k + n_2^k\|^2]$. From (34), we have:

$$\begin{aligned}
 \mathbb{E} \|\tilde{g}^k\|^2 &\leq 4C_1^2(d\sigma_1^2 + 1) \|x^k - w^k\|^2 + 4C_2^2(d\sigma_2^2 + 1) \|\nabla f(w^{k-1})\|^2 \\
 &\leq 4C^2(d\sigma_1^2 + 1) + 4G^2C_2^2(d\sigma_2^2 + 1) := \tilde{G}
 \end{aligned} \tag{47}$$

We now discuss a lower bound for $\mathbb{E}_k \langle \nabla f(x^k), g^k \rangle$. Our approach is to use the gradient sampling noise as a bridge to precisely characterize each term. Let $\Delta_i^k := \nabla f_i(x^k) - \nabla f_i(w^k)$, $\Delta^k := \nabla f(x^k) - \nabla f(w^k)$, and $\xi_{1i}^k := \Delta_i^k - \Delta^k$, $\xi_{2i}^k := \nabla f_i(w^k) - \nabla f(w^k)$. Then, we obtain:

$$\begin{aligned}
 \mathbb{E}_k[g^k] &= \mathbb{E}_k \left[\frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k), C_{1k}) + \frac{1}{|D|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k), C_{2k}) \right] \\
 &= \mathbb{E}_k \left[\Delta_i^k \cdot \min\left\{1, \frac{C_{1k}}{\|\Delta_i^k\|}\right\} \right] + \mathbb{E}_k \left[\nabla f_i(w^k) \cdot \min\left\{1, \frac{C_{2k}}{\|\nabla f_i(w^k)\|}\right\} \right] \\
 &= \mathbb{E}_k \left[(\Delta^k + \xi_{1i}^k) \cdot \min\left\{1, \frac{C_{1k}}{\|\Delta^k + \xi_{1i}^k\|}\right\} \right] + \mathbb{E}_k \left[(\nabla f(w^k) + \xi_{2i}^k) \cdot \min\left\{1, \frac{C_{2k}}{\|\nabla f(w^k) + \xi_{2i}^k\|}\right\} \right]
 \end{aligned} \tag{48}$$

Therefore, for $\mathbb{E}_k \langle \nabla f(x^k), g^k \rangle$, we have:

$$\begin{aligned}
 \mathbb{E}_k \langle \nabla f(x^k), g^k \rangle &= \langle \nabla f(x^k), \mathbb{E}_k[(\Delta^k + \xi_{1i}^k) \cdot \min\{1, \frac{C_{1k}}{\|\Delta^k + \xi_{1i}^k\|}\}] \\
 &\quad + \mathbb{E}_k[(\nabla f(w^k) + \xi_{2i}^k) \cdot \min\{1, \frac{C_{2k}}{\|\nabla f(w^k) + \xi_{2i}^k\|}\}] \rangle \\
 &= \underbrace{\langle \nabla f(x^k), \mathbb{E}_k[(\Delta^k + \xi_{1i}^k) \cdot \min\{1, \frac{C_{1k}}{\|\Delta^k + \xi_{1i}^k\|}\}] - \Delta^k \rangle}_{C:=Coupled Term} \\
 &\quad + \underbrace{\langle \nabla f(x^k), \mathbb{E}_k[(\nabla f(w^k) + \xi_{2i}^k) \cdot \min\{1, \frac{C_{2k}}{\|\nabla f(w^k) + \xi_{2i}^k\|}\}] - \nabla f(w^k) + \nabla f(x^k) \rangle}_{A:=Anchor Term}
 \end{aligned} \tag{49}$$

We denote $\mathbb{P}_1^k := \mathbb{E}_k[1_{\{\|\Delta^k + \xi_{1i}^k\| \leq C_{1k}\}}]$, $\mathbb{P}_2^k := \mathbb{E}_k[1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}}]$, and assume that $C_1 > 1$ and $C_2 > 1$. We then examine the two terms separately. First, for the term **A**, we have:

$$\begin{aligned}
 \mathbb{E}_k[(\nabla f(w^k) + \xi_{2i}^k) \cdot \min\{1, \frac{C_{2k}}{\|\nabla f(w^k) + \xi_{2i}^k\|}\}] &= \\
 \mathbb{E}_k[(\nabla f(w^k) + \xi_{2i}^k) \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}}] &+ \mathbb{E}_k[\frac{C_{2k} \cdot (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}}]
 \end{aligned} \tag{50}$$

Substituting into (49), we obtain:

$$\begin{aligned}
 A &= \|\nabla f(x^k)\|^2 + \langle \nabla f(x^k), -\nabla f(w^k) + \mathbb{E}_k[(\nabla f(w^k) + \xi_{2i}^k) \cdot \min\{1, \frac{C_{2k}}{\|\nabla f(w^k) + \xi_{2i}^k\|}\}] \rangle \\
 &= \|\nabla f(x^k)\|^2 + \langle \nabla f(x^k), -\nabla f(w^k) + \mathbb{E}_k[(\nabla f(w^k) + \xi_{2i}^k) \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}}] \\
 &\quad + \mathbb{E}_k[\frac{C_{2k} \cdot (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}}] \rangle \\
 &= \|\nabla f(x^k)\|^2 + \mathbb{E}_k[\langle \nabla f(x^k), -\nabla f(w^k) \cdot (1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}} + 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}}) \\
 &\quad + (\nabla f(w^k) + \xi_{2i}^k) \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}} + \frac{C_{2k} \cdot (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}}] \rangle \\
 &= \|\nabla f(x^k)\|^2 + \mathbb{E}_k[\langle \nabla f(x^k), \xi_{2i}^k \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}} \rangle] \\
 &\quad + \mathbb{E}_k[\langle \nabla f(x^k), [\frac{C_{2k} \cdot (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} - f(w^k)] \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle]
 \end{aligned} \tag{51}$$

Focusing on the final term of (51) alone, we obtain:

$$\begin{aligned}
 &\mathbb{E}_k[\langle \nabla f(x^k), [\frac{C_{2k} \cdot (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} - f(w^k)] \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle] \\
 &= \mathbb{E}_k[\langle \nabla f(x^k), [\frac{C_2 \|\nabla f(w^{k-1})\| (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} - (f(w^k) + \xi_{2i}^k) + \xi_{2i}^k] \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle] \\
 &= \mathbb{E}_k[\langle \nabla f(x^k), [\frac{[C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\|] (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} + \xi_{2i}^k] \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle] \\
 &= \mathbb{E}_k[\langle \nabla f(x^k), [\frac{[C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\|] (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle] \\
 &\quad + \mathbb{E}_k[\langle \nabla f(x^k), \xi_{2i}^k \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle]
 \end{aligned} \tag{52}$$

Substituting into (51), we obtain:

$$\begin{aligned}
 A &= \|\nabla f(x^k)\|^2 + \mathbb{E}_k[\langle \nabla f(x^k), \xi_{2i}^k \cdot (1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}} + 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}}) \rangle] \\
 &\quad + \mathbb{E}_k[\langle \nabla f(x^k), \frac{[C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\|] (\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \rangle]
 \end{aligned} \tag{53}$$

Since $1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| \leq C_{2k}\}} + 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} = 1$, and $C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\| \geq C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k)\| - \|\xi_{2i}^k\|$, together with the facts that $\mathbb{E}_k[\xi_{2i}^k] = 0$ and $(C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\|) \cdot 1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \leq 0$, we can, by applying the Cauchy inequality, derive a lower bound for $\mathbb{E}[A]$:

$$\begin{aligned}
 \mathbb{E}[A] &= \mathbb{E} \|\nabla f(x^k)\|^2 \\
 &+ \mathbb{E} \left[\underbrace{\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot [C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\|] \cdot \langle \nabla f(x^k), \frac{(\nabla f(w^k) + \xi_{2i}^k)}{\|\nabla f(w^k) + \xi_{2i}^k\|} \rangle]}_{<0} \right] \\
 &\stackrel{(a)}{\geq} \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot [C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k) + \xi_{2i}^k\|] \cdot \|\nabla f(x^k)\|] \right] \\
 &\stackrel{(b)}{\geq} \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot [C_2 \|\nabla f(w^{k-1})\| - \|\nabla f(w^k)\| - \|\xi_{2i}^k\|] \cdot \|\nabla f(x^k)\|] \right] \\
 &\geq \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot [C_2 \|\nabla f(w^k)\| \right. \\
 &\quad \left. - C_2 \|\nabla f(w^k) - \nabla f(w^{k-1})\| - \|\nabla f(w^k)\| - \|\xi_{2i}^k\|] \cdot \|\nabla f(x^k)\|] \right] \\
 &\stackrel{(c)}{\geq} \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot [(C_2 - 1) \|\nabla f(w^k)\| - \|\xi_{2i}^k\|] \cdot \|\nabla f(x^k)\|] \right. \\
 &\quad \left. - \mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} L \|w^k - w^{k-1}\| \cdot \|\nabla f(x^k)\|] \right] \\
 &\stackrel{(d)}{\geq} \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[(1 - \mathbb{P}_2^k)(C_2 - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \right] - \mathbb{E} \left[\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot \|\xi_{2i}^k\| \cdot \|\nabla f(x^k)\|] \right. \\
 &\quad \left. - L p \mathbb{E} [\|x^{k-1} - w^{k-1}\| \cdot \|\nabla f(x^k)\|] \right] \\
 &\stackrel{(e)}{\geq} \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[(1 - \mathbb{P}_2^k)(C_2 - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \right] - \mathbb{E} \left[\mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot \|\xi_{2i}^k\| \cdot \|\nabla f(x^k)\|] \right] \\
 &\quad - \eta \cdot LG\tilde{G}
 \end{aligned} \tag{54}$$

Here, (a) follows from the Cauchy inequality; (b) follows from the triangle inequality; (c) follows from the L-smooth property; (d) comes from the iterative update of w^k and (e) follows directly from (33). We now consider the third term in the above expression. Let S_k denote the set of ξ_{2i}^k such that $\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}$, and define $P_{k,z} := \Pr(\xi_{2i}^k \in S_k, \|\xi^k\| = z)$. Then, we have:

$$\begin{aligned}
 &- \mathbb{E}_k [1_{\{\|\nabla f(w^k) + \xi_{2i}^k\| > C_{2k}\}} \cdot \|\xi_{2i}^k\| \cdot \|\nabla f(x^k)\|] \\
 &= - \mathbb{E}_k [1_{\{\xi^k \in S_k\}} \cdot \|\xi^k\| \cdot \|\nabla f(x^k)\|] \\
 &= - \|\nabla f(x^k)\| \cdot \int_0^{+\infty} P_{k,z} \cdot z \, dz \\
 &= - \|\nabla f(x^k)\| \cdot \int_0^{+\infty} \sqrt{P_{k,z}} \cdot \sqrt{z^2 \cdot P_{k,z}} \, dz \\
 &\geq - \|\nabla f(x^k)\| \cdot \sqrt{\left(\int_0^{+\infty} P_{k,z} \, dz \right) \cdot \left(\int_0^{+\infty} z^2 P_{k,z} \, dz \right)} \\
 &\geq - \|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_2^k)} \cdot \sqrt{\mathbb{E}_k [\|\xi_{2i}^k\|^2]}
 \end{aligned} \tag{55}$$

Thus, we can obtain:

$$\mathbb{E}[A] \geq \mathbb{E} \|\nabla f(x^k)\|^2 + \mathbb{E} \left[(1 - \mathbb{P}_2^k)(C_2 - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \right] - \mathbb{E} \left[\|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_2^k)} \cdot \sqrt{\mathbb{E}_k [\|\xi_{2i}^k\|^2]} \right] - \eta \cdot LG\tilde{G} \tag{56}$$

The treatment of C is analogous to that of A . Due to the symmetry between $\nabla f(w^k)$ and Δ^k in the expressions for A and

C , and by referring to (49)–(56), we can obtain:

$$\begin{aligned}
 C &= \langle \nabla f(x^k), \mathbb{E}_k \left[\frac{\Delta^k + \xi_{1i}^k}{\|\Delta^k + \xi_{1i}^k\|} \cdot \underbrace{(C_{1k} - \|\Delta^k + \xi_{1i}^k\|) \cdot 1_{\{\|\Delta^k + \xi_{1i}^k\| > C_{1k}\}}}_{<0} \right] \rangle \\
 &\geq \mathbb{E}_k[(C_{1k} - \|\Delta^k + \xi_{1i}^k\|) \cdot 1_{\{\|\Delta^k + \xi_{1i}^k\| > C_{1k}\}}] \cdot \|\nabla f(x^k)\| \\
 &\geq \mathbb{E}_k[(C_1 \|x^k - w^k\| - \|\Delta^k\| - \|\xi_{1i}^k\|) \cdot 1_{\{\|\Delta^k + \xi_{1i}^k\| > C_{1k}\}}] \cdot \|\nabla f(x^k)\| \\
 &\geq \mathbb{E}_k[(\frac{C_1}{L} - 1) \|\Delta^k\| \cdot 1_{\{\|\Delta^k + \xi_{1i}^k\| > C_{1k}\}}] \cdot \|\nabla f(x^k)\| \\
 &\quad - \mathbb{E}_k[\|\nabla f(x^k)\| \cdot \|\xi_{1i}^k\| \cdot 1_{\{\|\Delta^k + \xi_{1i}^k\| > C_{1k}\}}] \\
 &\geq (1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1) \|\Delta^k\| \cdot \|\nabla f(x^k)\| - \|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_1^k)} \cdot \sqrt{\mathbb{E}_k[\|\xi_{1i}^k\|^2]} \\
 &\geq (1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1) \|\nabla f(x^k) - \nabla f(w^k)\| \cdot \|\nabla f(x^k)\| \\
 &\quad - \|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_1^k)} \cdot \sqrt{\mathbb{E}_k[\|\xi_{1i}^k\|^2]}
 \end{aligned} \tag{57}$$

By Assumption 3.4, we have $\mathbb{E}_k[\|\xi_{2i}^k\|^2] = \text{Var}(\nabla f_i) \leq \tau^2$. Moreover, $\xi_{1i}^k = \Delta_i^k - \Delta^k = \nabla f_i(x^k) - \nabla f_i(w^k) - \nabla f(x^k) + \nabla f(w^k) = (\nabla f_i(x^k) - \nabla f(x^k)) - (\nabla f_i(w^k) - \nabla f(w^k)) = \xi_{2i}^k - \xi_{2i}^{k'}$, which implies $\mathbb{E}_k[\|\xi_{1i}^k\|^2] \leq 4\tau^2$. Combining the results from both terms, we obtain:

$$\begin{aligned}
 \mathbb{E}[\langle \nabla f(x^k), g^k \rangle] &\geq \mathbb{E}[\|\nabla f(x^k)\|^2] + \mathbb{E}(1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \\
 &\quad + \mathbb{E}(1 - \mathbb{P}_2^k)(C_2 - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| - 2\mathbb{E}[\|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_1^k)} \cdot \tau] \\
 &\quad - \mathbb{E}[\|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_2^k)} \cdot \tau - \eta \cdot LG\tilde{G}]
 \end{aligned} \tag{58}$$

Below, we consider two cases, namely $\|\nabla f(x^k)\| \geq 3\tau$ and $\|\nabla f(x^k)\| < 3\tau$, and we will use probabilities to combine them. For the former case, we have:

$$\begin{aligned}
 \mathbb{E}[\langle \nabla f(x^k), g^k \rangle] + LG\tilde{G}\eta &\geq (1 - \frac{2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}}{3}) \|\nabla f(x^k)\|^2 \\
 &\quad + \frac{2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}}{3} \|\nabla f(x^k)\|^2 \\
 &\quad - (2\sqrt{(1 - \mathbb{P}_1^k)} + \sqrt{(1 - \mathbb{P}_2^k)}) \|\nabla f(x^k)\| \tau \\
 &\quad + (1 - \mathbb{P}_2^k)(C_2 - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \\
 &\quad + (1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \\
 &\geq (1 - \frac{2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}}{3}) \|\nabla f(x^k)\|^2 \\
 &\quad + (1 - \mathbb{P}_2^k)(C_2 - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \\
 &\quad + (1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1) \|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \geq 0
 \end{aligned} \tag{59}$$

In summary, by combining the two cases using probabilities, let $\mathbb{P}^k := \Pr(\|\nabla f(x^k)\| < 3\tau \mid x^{k-1})$. Then, we have:

$$\begin{aligned}
 \mathbb{E}[\langle \nabla f(x^k), g^k \rangle] + LG\tilde{G}\eta &\geq \mathbb{E}(1 - \mathbb{P}^k)(1 - \frac{2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}}{3})\|\nabla f(x^k)\|^2 \\
 &\quad + \mathbb{E}(1 - \mathbb{P}^k)(1 - \mathbb{P}_2^k)(C_2 - 1)\|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| \\
 &\quad + \mathbb{E}(1 - \mathbb{P}^k)(1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1)\|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \\
 &\quad + \mathbb{E}[\mathbb{P}^k\|\nabla f(x^k)\|^2] + \mathbb{E}[\mathbb{P}^k(1 - \mathbb{P}_1^k)(\frac{C_1}{L} - 1)\|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\|] \\
 &\quad + \mathbb{E}[\mathbb{P}^k(1 - \mathbb{P}_2^k)(C_2 - 1)\|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\|] - \mathbb{E}[2\mathbb{P}^k\|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_1^k)} \cdot \tau] \\
 &\quad - \mathbb{E}[\mathbb{P}^k\|\nabla f(x^k)\| \cdot \sqrt{(1 - \mathbb{P}_2^k)} \cdot \tau] \\
 &= \underbrace{\mathbb{E}(1 - (1 - \mathbb{P}^k)\frac{2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}}{3})\|\nabla f(x^k)\|^2}_{0 \leq \lambda_1^k \leq 1} \\
 &\quad + \underbrace{\mathbb{E}(1 - \mathbb{P}^k)(C_2 - 1)\|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\|}_{\lambda_2^k \geq 0} \\
 &\quad + \underbrace{\mathbb{E}(1 - \mathbb{P}^k)(\frac{C_1}{L} - 1)\|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\|}_{\lambda_3^k \geq 0} \\
 &\quad - \underbrace{\mathbb{E}[\mathbb{P}^k \cdot \frac{2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}}{3} \cdot \|\nabla f(x^k)\| \cdot 3\tau]}_{0 \leq \lambda_4^k \leq 1}
 \end{aligned} \tag{60}$$

For notational convenience, we further define and restate:

$$\begin{aligned}
 \lambda_1^k &:= 1 - \frac{1}{3}(1 - \mathbb{P}^k)(2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}), \quad \lambda_2^k := (1 - \mathbb{P}^k)(C_2 - 1), \quad \lambda_3^k := (1 - \mathbb{P}^k)(\frac{C_1}{L} - 1) \\
 \lambda_4^k &:= \frac{1}{3}\mathbb{P}^k(2\sqrt{1 - \mathbb{P}_1^k} + \sqrt{1 - \mathbb{P}_2^k}), \quad \mathbb{P}^k := \Pr(\|\nabla f(x^k)\| \leq 3\tau \mid x^{k-1}), \\
 \mathbb{P}_1^k &:= \mathbb{E}_k[1_{\{\|\nabla f_i(x^k) - \nabla f_i(w^k)\|_2 \leq C_{1k}\}}], \quad \mathbb{P}_2^k := \mathbb{E}_k[1_{\{\|\nabla f_i(w^k)\|_2 \leq C_{2k}\}}].
 \end{aligned}$$

By substituting (60), (47) into (46), taking the full expectation on both sides of the inequality, summing over $k = 1$ to T , and setting $\eta = \sqrt{\frac{2(f(x^0) - f(x^*))}{TL\tilde{G}(1+4G)}}$, we obtain:

$$\begin{aligned}
 &\frac{1}{T} \sum_{k=1}^T \mathbb{E} \left[\lambda_1^k \|\nabla f(x^k)\|^2 + \lambda_2^k \|\nabla f(x^k)\| \cdot \|\nabla f(w^k)\| + \lambda_3^k \|\nabla f(x^k)\| \cdot \|\nabla f(x^k) - \nabla f(w^k)\| \right] \\
 &\leq \frac{f(x^0) - f(x^*)}{\eta T} + \frac{\eta L}{2T} \sum_{k=1}^T \tilde{G} + \frac{2\eta}{T} \sum_{k=1}^T LG\tilde{G} + \frac{1}{T} \sum_{k=1}^T \mathbb{E} \left[\lambda_4^k \cdot 3\tau \|\nabla f(x^k)\| \right] \\
 &\leq 2\sqrt{\frac{(f(x^0) - f(x^*))L\tilde{G}(1+4G)}{2T}} + \frac{1}{T} \sum_{k=1}^T \mathbb{E} \left[\lambda_4^k \cdot 3\tau \|\nabla f(x^k)\| \right]
 \end{aligned} \tag{61}$$

E. Additional Experiments

In this section, we provide additional information and results on our numerical experiments that are not given in the main paper due to the space limitation.

Datasets Information We conduct experiments on Mushroom, MNIST, CIFAR-10, CIFAR-100, IMDB, and GLUE-SST-2. The information of all datasets used is summarized in Table.5.

Table 5. The summary of the datasets used in the experiments.

Dataset	Samples	Type	Classes	Task
Mushroom	8,124	Tabular	2	SVM
MNIST	70,000	Image (28×28, Gray)	10	CV
CIFAR-10	60,000	Image (32×32, RGB)	10	CV
CIFAR-100	60,000	Image (32×32, RGB)	100	CV
IMDb	50,000	Text (Reviews)	2	NLP
GLUE-SST-2	67,349	Text (Sentences)	2	NLP

Results on Different C_1 and C_2 First, we provide an ablation study on the selection of clipping thresholds C_1 and C_2 . We conduct experiments on the CIFAR-10 dataset with the learning rate set to $\eta = 0.0125$ and the privacy parameter $(\epsilon, \delta) = (5, 10^{-5})$. Following the main experiment, we set the mini-batch size to $|S| = 256$, the large-batch size to $|D'| = 4096$ and $p = \frac{2|S|}{|D'|} = 0.125$. We fix $C = 1$ and vary C_1 and C_2 over the range $\{0.125, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64\}$. We report the results for each configuration and compare them against DP-SGD. The experimental results of DP-C4 is presented in Figure.2. In each cell of the heatmap, the color encodes the corresponding accuracy, with warmer shades indicating higher accuracy and cooler shades indicating lower accuracy. Each cell further reports the accuracy associated with the corresponding clipping thresholds, while the value in parentheses denotes the accuracy difference relative to DP-SGD.

On the one hand, for DP-C4, when examining a single row or column of the grid, we observe that increasing C_1 initially improves accuracy, which subsequently decreases; a similar trend is observed when increasing C_2 . More specifically, as C_1 and C_2 gradually increase, the injected noise becomes larger, leading to a gradual degradation in accuracy until it converges to a constant value. In particular, when $C_1 = C_2 = 64$, the accuracies of DP-C4 converge to 61.16, since in this case a constant clipping threshold is applied at each iteration (i.e., in DP-C4: $C_{1k} = \min\{C, C_1\|x^k - w^k\|\} = C$, $C_{2k} = \min\{C, C_2\|\nabla f(w^{k-1})\|\} = C$).

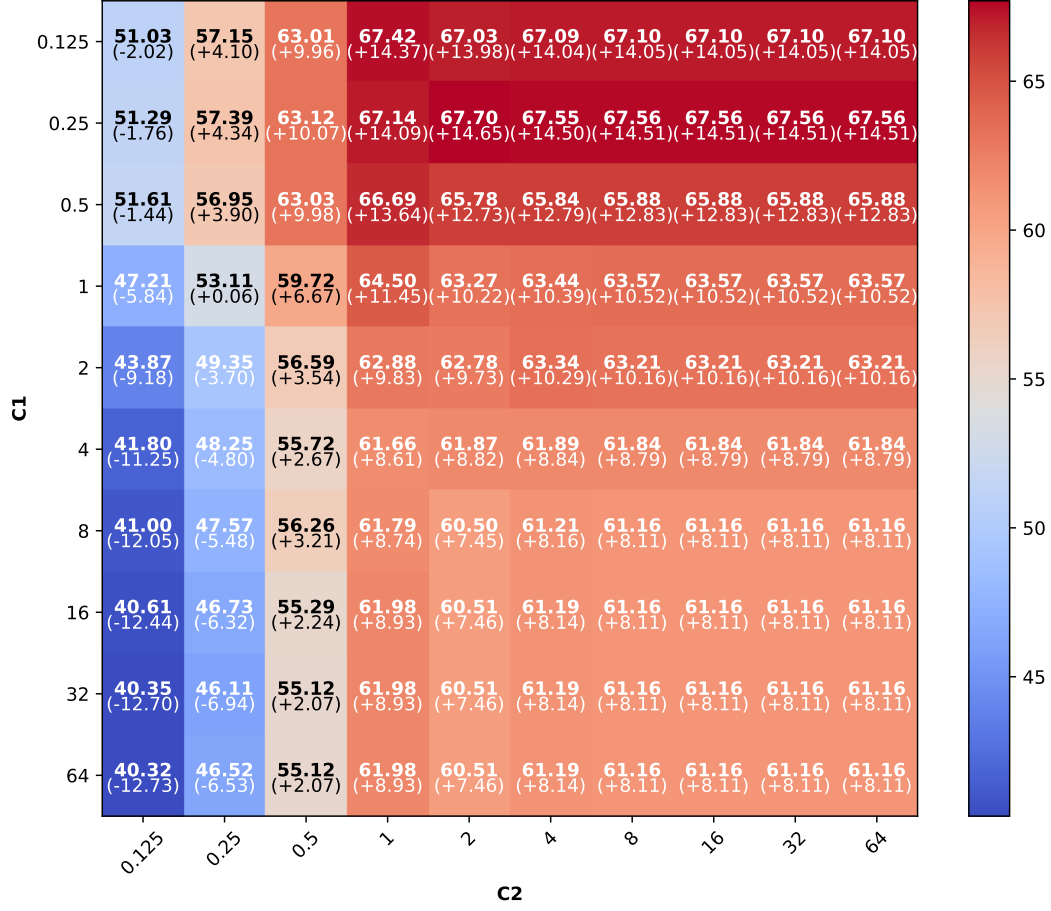
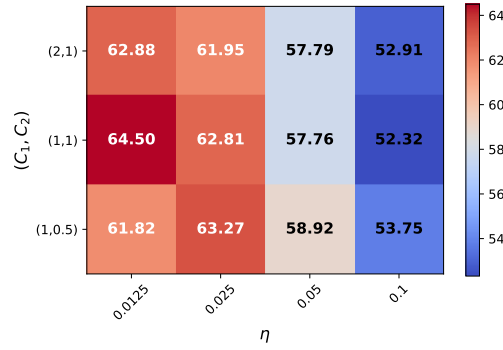
On the other hand, when C_1 is small, the accuracy does not decrease significantly. This is because as $C_1 \rightarrow 0$, the coupled term of DP-C4 vanishes, which essentially reduces the method to a large-batch variant of delayed DP-SGD. The iterative structure is thus not severely disrupted, while the injected noise is substantially reduced. In contrast, when C_2 is small, accuracy drops more sharply. This is due to the excessive clipping bias, which prevents effective updates (i.e., $\frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k)) + \frac{1}{|D'|} \sum_{i \in D} \text{clip}(\nabla f_i(w^k)) \approx \frac{1}{|S|} \sum_{i \in S} \text{clip}(\nabla f_i(x^k) - \nabla f_i(w^k))$). In summary, the vanishing of the coupled term can be tolerated since it still preserves an effective optimization structure, whereas the vanishing of the anchor term is detrimental, as it leads to severe performance degradation.

Results on Different C We also conduct an ablation study on the overall clipping threshold C . The experiments are performed on CIFAR-10 with $\eta = 0.0125$, $|S| = 256$, $|D'| = 4096$, the privacy parameter $(\epsilon, \delta) = (5, 10^{-5})$, and $p = 0.125$. We fix $C_1 = C_2 = 1$ and vary C over the set $\{0.125, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64\}$. The results comparing DP-C4 with DP-SGD are summarized in Table 6.

 Table 6. Test accuracy of different methods on different clipping threshold C .

Method	Values of Clipping Threshold C									
	0.125	0.25	0.5	1	2	4	8	16	32	64
DP-SGD	54.91	55.36	53.42	53.05	41.01	27.40	18.36	16.41	14.93	10.74
DP-C4	55.84	59.22	61.52	64.50	61.30	56.38	55.71	55.25	55.25	55.25

It can be observed that, on the one hand, as C decreases, the accuracy of both DP-SGD and DP-C4 first increases and then decreases. This behavior is attributed to the reduction of the injected noise and the simultaneous growth of the clipping bias. When C becomes sufficiently small, every term is clipped on a per-sample basis, and thus the iterations of both methods resemble a normalized update scheme. On the other hand, as C increases, the accuracy of DP-SGD drops rapidly, while that of DP-C4 decreases more slowly and eventually converges to a fixed accuracy level of 55.25%. This robustness stems from the fact that the effective clipping thresholds of DP-C4 are determined by $C_{1k} = \min\{C, C_1\|x^k - w^k\|\} \leq \frac{2C_1G}{L}$, $C_{2k} = \min\{C, C_2\|\nabla f(w^{k-1})\|\} \leq C_2G$, which are governed by the gradient


 Figure 2. Accuracy of DP-C4 with different C_1 and C_2

 Figure 3. Accuracy of DP-C4 with different η

norm, therefore do not grow unbounded.

Results on Different η We conduct experiments on CIFAR-10 using DP-C4 under different η , while keeping the remaining parameters unchanged, and record the corresponding accuracies of DP-C4. The detailed results are presented in Figure.3.

Results on Different Routines We further conduct experiments on CIFAR-10 using different routines. We fix $C = C_1 = C_2 = 1$, while keeping the remaining parameters unchanged. The results are reported in Table 7.

Table 7. Test accuracy of DP-C4 on different routines.

Method	Different Routines			
	1	2	3	4
DP-C4	64.50	64.39	63.97	64.28

We observe that the results of the four routines are similar. This is because the different routines only modify the update strategy of w^k and do not alter the intrinsic properties of the DP-C4 iterative scheme, so that their behavior is highly similar in expectation.

Results on Different Large Batchesizes We conduct experiments on CIFAR-10 using DP-C4 under different large-batch sizes. The learning rate is set to $\eta = 0.0125$, with $C = C_1 = C_2 = 1$, $|S| = 256$, and $p = 0.125$. We vary the large-batch size as $|D'| \in \{512 = 2 \cdot |S|, 2^2 \cdot |S|, 2^3 \cdot |S|, 2^4 \cdot |S|, 2^5 \cdot |S| = 8192\}$, and record the corresponding accuracies of DP-C4. The detailed results are presented in Table 8.

Table 8. Test accuracy of DP-C4 on different large-batch sizes.

Method	Different Large-batch Sizes				
	512	1024	2048	4096	8192
DP-C4	41.12	52.99	60.98	64.50	58.91

We observe that as $|D'|$ increases, the accuracy of DP-C4 first rises and then decreases. This behavior occurs because a relatively small large batch leads to inaccurate estimation of the full gradient and, compared to DP-SGD, introduces excessive clipping bias. Conversely, an excessively large batch significantly increases the number of samples averaged in each iteration, which effectively reduces the number of updates and consequently degrades performance.

Results on Different p We conducted experiments on CIFAR-10 to evaluate DP-C4 under different update probabilities p . We set the learning rate to $\eta = 0.0125$, with $C = C_1 = C_2 = 1$, $|S| = 256$, and $|D'| = 4096$. We varied $p \in \{\frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \frac{1}{2^5}\}$ and recorded the corresponding accuracy of DP-C4. The detailed results are presented in Table 9.

Table 9. Test accuracy of DP-C4 on different p .

Method	Different p				
	0.5	0.25	0.125	0.0625	0.03125
DP-C4	60.04	63.67	64.50	63.65	62.71

We can observe that as p decreases, the accuracy of DP-C4 first increases and then decreases. This phenomenon can be explained as follows: when p is relatively large, the anchor term is updated frequently, which increases the average data consumption per iteration and consequently reduces the effective number of iterations, leading to suboptimal performance. On the other hand, when p is too small, the anchor term is updated too infrequently, which also negatively impacts the accuracy.

Results on C_{1k} and C_{2k} across Iterations We evaluate the effectiveness of the clipping strategy in DP-C4 for noise control on CIFAR-10. We set the learning rate to $\eta = 0.0125$, $|S| = 256$, $|D'| = 4096$, $p = 0.125$. Under three configurations—fully balanced, imbalanced ($C_1 \neq C_2$), and large- C —we track C_{1k} and C_{2k} at every iteration over the first 20 training epochs. It can be observed that during the optimization process, for a subset of iterations k , both C_{1k} and C_{2k} are smaller than the preset clipping bound C , which effectively limits the magnitude of the injected noise. The detailed results are presented in Figure.4.

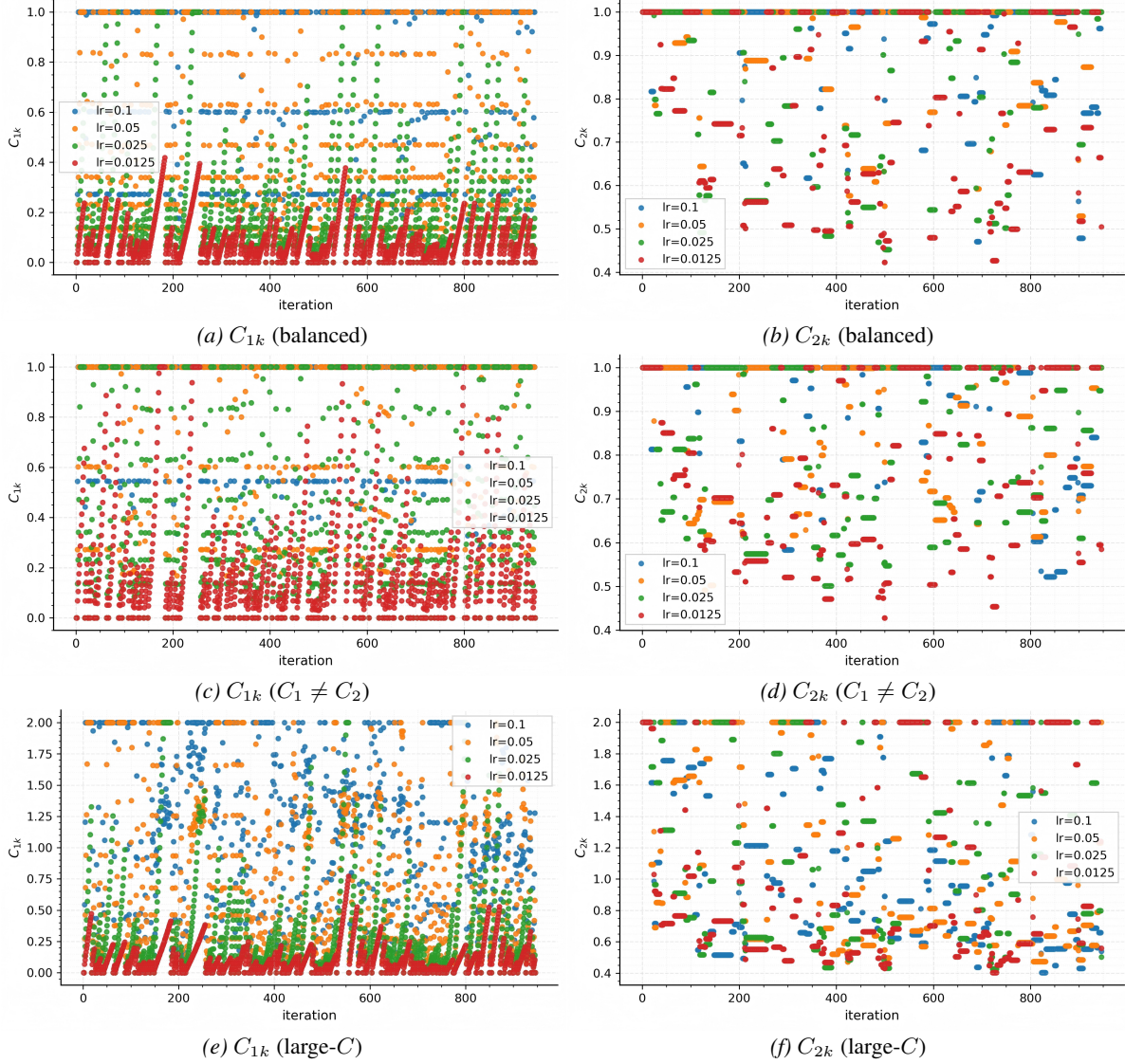


Figure 4. Iteration-wise evolution of C_{1k} and C_{2k} under three clipping regimes on CIFAR-10. Each row corresponds to a different configuration: balanced ($C_1 = C_2 = C = 1$), imbalanced ($C_1 = 2, C_2 = C = 1$), and larger- C ($C_1 = C_2 = 1, C = 2$).